

Effects of Lit and Dark Market Fragmentation on Liquidity

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Abstract

Based on data from eight exchanges and a trade reporting facility for a large sample of LSE- and Euronext-listed equities, this article investigates how lit and dark market fragmentation affects liquidity. Neither dark trading, nor fragmentation between lit order books, is found to harm liquidity. Lit fragmentation improves spreads and depth across markets and locally on the primary exchange, or at worst does not affect them. Benefits are greater for large stocks and stocks with less electronic trading. Lit fragmentation however harms the depth of small stocks. Adverse effects on the depth of large stocks result from algorithmic trading and not from fragmentation.

1. Introduction

With the development of sophisticated trading technologies and the enforcement of pro-competition market regulations, fragmentation of trading volumes between competing trading venues has undeniably increased in all large western stock markets. More than fifteen trading venues are now active in the U.S., with more than 45% of volume in S&P 500 stocks and around 55% of volume in Nasdaq-100 stocks traded off primary exchanges in 2016 according to Fidessa¹. In Europe, more than ten trading venues have now become fairly active in liquid European stocks. Levels of fragmentation in stock markets however strongly differ across countries and exhibit time-series variations. On average, the level of fragmentation in U.S. stock markets is twice that observed in main European marketplaces and in the Canadian market, while at the opposite, stock trading is relatively concentrated on the primary exchange in Japan. It is thus important to understand how those differences impact trading costs, and thereby investment performance, for domestic and international investors.

A market is fragmented when trading takes place simultaneously at different locations. The trade flow in a security may fragment in two ways essentially: (1) several open and transparent order books compete for the order flow; (2) a portion of the order flow is internalized off-exchange by dealers or crossing engines, either under regulation or in the over-the-counter (OTC) market. Case (1) will be referred to as lit fragmentation while case (2) will be referred to as dark fragmentation or dark trading.

While there is concern among regulators, issuers, and asset managers, about how the different forms of fragmentation and their level may impact the quality of order execution and market liquidity, the empirical evidence on this issue is relatively scarce, probably because obtaining

¹ Fidessa is a private company which provides trading systems, market data, and connectivity solutions.

exhaustive data from all active markets on large stock samples is a difficult hurdle to overcome. The present article complements the literature by examining how the consolidated liquidity of competing trading systems, here called global liquidity, as well as the local liquidity of the primary exchange, relate to lit and dark fragmentation, on a large sample of equities listed at the London Stock Exchange (LSE) and on Euronext. To my knowledge, it is the first paper to address the issue with both a long time series and a large sample. An original aspect of the study lies in that it is conducted with two distinct approaches: a first approach which treats the shift from consolidated to fragmented markets legally supported by the implementation of the Markets in Financial Instruments Directive (MiFID)² as a natural experiment, and a second approach which investigates the time series of spreads, depth, and fragmentation measures in fragmented markets. Both approaches provide novel insights by disentangling the effects of algorithmic trading (AT) and fragmentation. They also provide new evidence on differentiated impacts of fragmentation according to market size and to the initial level of automation and/or centralization of the order flow.

Results show that (1) spreads and depth improve with, or at worst are not affected by, lit fragmentation after controlling for endogeneity and AT; (2) when both AT and lit fragmentation contribute to improving spreads, the effect of fragmentation is economically more significant; (3) stocks with less electronic trading before MiFID benefit more from multiple-venue trading; (4) dark trading does not harm liquidity but may, on the contrary, improve some of its dimensions; (5) lit fragmentation reduces the depth of smaller stocks but adverse effects on the depth of large stocks result from AT and not from fragmentation.

² This European Union law provides a comprehensive and harmonized regulatory regime for investment services and activities across the 30 member states of the European Economic Area (EEA), i.e. the 27 member states of the European Union (EU) plus Norway, Iceland, and Liechtenstein.

The remainder of the paper is organized as follows. Section 2 reviews the related literature and presents the research objectives. Section 3 provides details about the regulatory framework and the organization of the marketplace. Section 4 describes the sample and the data. The fragmentation and liquidity measures used in the empirics are presented in Section 5. Sections 6 and 7 are dedicated to comparing the liquidity of the pre-MiFID quasi-consolidated marketplace with that of post-MiFID fragmented markets. Section 8 exposes panel analyses of the relation between fragmentation and liquidity in the post-MiFID fragmented markets. Section 9 concludes.

2. Literature review and research objectives

Lit fragmentation, i.e., fragmentation between transparent trading platforms, is generally considered as producing positive competition effects (Hamilton, 1979; Stoll, 2003). Bid-ask spreads have been observed to narrow in the incumbent market for diverse financial instruments when a new market starts operating (Battalio, 1997; Huang, 2002; Mayhew, 2002; Boehmer and Boehmer, 2003; Fontnouvelle et al., 2003; Nguyen, Van Ness, and Van Ness, 2007). Foucault and Menkveld (2008) also show that, due to the absence of price priority across markets, consolidated depth is larger after the entry of a new order book. Nevertheless, market fragmentation may have adverse effects. First, the dispersion of trading may “prevent full realization of any economies of centralized trading on the exchange” (Hamilton, 1979, p.186). Second, fragmentation may harm liquidity by increasing information asymmetry costs. Chowdry and Nanda (1991) demonstrate that adverse selection costs increase with the number of markets listing an asset. Easley et al. (1996) and Bessembinder and Kaufman (1997) also coin that when a new market opens for a stock, it may skim the least informed and consequently most profitable orders, and then harm the liquidity of the primary market. Empirically, those adverse effects are

usually observed when dealers fragment the market. Bennett and Wei (2006) provide empirical evidence that stocks switching from a fragmented market of dealers, as the old Nasdaq structure, to a more consolidated structure such as the electronic order book of NYSE, experience an improvement in spreads.

While most of this literature provides evidence on the benefits of competition between markets, none of the cited papers analyzes the direct interaction between fragmentation and liquidity when an asset trades on several venues. The first empirical study to address this issue is that of O'Hara and Ye (2011), who exploited Securities and Exchange Commission (SEC) Rule 605 monthly data for 262 US stocks over six months of 2008. They find that high fragmented stocks have lower transaction costs and faster execution speed than low fragmented stocks, but their approach is essentially a cross-sectional analysis and cannot distinguish cross-market consolidated liquidity from local liquidity in a given market due to the nature of the data exploited. Degryse et al. (2015) are able to make this distinction with detailed order book data for 52 Dutch stocks. They find that lit fragmentation improves the depth of the consolidated marketplace but reduces that of the primary exchange. While their work makes a valuable contribution thanks to a very long time-series analysis and a sound methodology, their sample only covers one country whose stock market is not among the largest in Europe, which raises the question of whether their results can be generalized.

As a result, to date, no work has investigated the link between lit fragmentation and liquidity with high-frequency data covering all trading venues for a large sample of equities primary listed on leading exchanges. My main research objective is to fill this gap by exploiting a large sample of European stocks listed on the LSE and on Euronext and by measuring liquidity for global traders who are connected to several platforms as well as for local traders whose trading universe

is restricted to the primary exchange. The market size diversity in the sample allows me to compare large and small stocks and to test how fragmentation effects on liquidity differs with market size. As adverse selection is greater on small stocks, adverse effects of fragmentation due to information asymmetry can be expected to be more pronounced for those stocks.

Regarding the impact of dark fragmentation on liquidity, the conclusions in the literature are diverse and differ according to the form of dark trading considered. One form of dark trading is when dealers or intermediaries internalize trades off-exchange. Human intermediaries are the most able to cream-skim profitable orders. Barclay et al. (2003) find evidence that market makers can preference or internalize the less informed trades. Using European data, Gajewski and Gresse (2007) show that off-book trading in a hybrid market tends to increase trading costs in comparison with trading in a centralized order book. Those findings suggest that the portion of dark trading corresponding to dealers' off-exchange executions may be more detrimental to liquidity than lit fragmentation. Dark pools running crossing engines are also likely to skim off the uninformed order flow. Zhu (2014) develops a model in which crossing networks (CNs) are proved to be more attractive to the uninformed. Under this hypothesis, dark trading should increase price informativeness but at the expense of greater adverse selection costs in the main market.

This view is challenged by a couple of papers. Buti et al. (2016a) demonstrate that spreads increase with CN trading only if the primary order book is thin, otherwise spreads narrow. Converse to Zhu (2014), Ye (2012) demonstrates that a proportion of informed traders migrate to CNs, which lessens price impact on-exchange and thereby reduces spreads. With private data from a CN, Gresse (2006) shows that CNs offers additional risk sharing capacities which contributes to improving liquidity in the primary market. Further, according to the empirical

work of Buti et al. (2016b), Brugler (2015), and Foley and Putniņš (2016), dark market activity improves liquidity.

Given the mixed theoretical and empirical findings on dark fragmentation in the literature, the second research objective of this paper is to test which of the positive and the negative effect of dark trading on liquidity dominates.

3. Stock universe, regulatory framework, and market organization

My empirical strategy consists of two approaches: (a) a natural experiment analysis around the implementation of MiFID on 1 November 2007 and (b) a time series analysis of the relation between fragmentation and liquidity in the post-MiFID world. The first approach consists of comparing liquidity in the pre-MiFID era when trading was concentrated on primary exchanges with the liquidity of the fragmented markets of the post-MiFID era. In the second approach, I analyze the two-way temporal relation between fragmentation and liquidity by a two-stage panel methodology over a later post-MiFID period. The impact of dark trading on liquidity can only be tested with the second empirical approach, as data on dark trading are only available in the post-MiFID period.

The stock universe I consider are the constituents of the FTSE 100 index, primary listed at the LSE, and the constituents of the CAC 40 and SBF 80 indices, primary listed on Euronext. The lit trading platforms considered for the empirics are those of the primary exchanges plus any other lit platform on which the sample stocks may trade. Dark trading considered in the study encompasses OTC trades as well as trades internalized by investment banks through regulated mechanisms.

3.1. Stock universe

The FTSE 100 is comprised of the largest capitalization stocks of the LSE. The CAC 40 is the flagship index of Euronext Paris and comprises the largest stocks of Euronext Paris as well as some large-capitalization stocks primarily listed on Euronext Amsterdam and Euronext Brussels. The SBF 80 index comprises 80 middle capitalization stocks listed on Euronext Paris and elected for being the most traded just after the CAC 40 components.³ Studying FTSE 100 and CAC 40 securities allows to compare stocks with different market structure on their primary exchange. Comparing the SBF 80 and the CAC 40 stocks allows to examine whether market fragmentation affects medium-capitalization stocks differently.

3.2. Regulatory framework

Until early 2007, secondary market trading in European stock markets was very concentrated on primary exchanges. In 2007 and 2008, a handful of pan-European trading platforms have emerged and grabbed significant market shares: BATS Europe, the subsidiary of U.S. exchange BATS, opened on 31 October 2008; Chi-X, owned by broker Instinet at the time of the study and then taken over by BATS Europe in 2011, began trading a few Dutch and German stocks on 30 March 2007, and then progressively extended its universe of stocks in 2007 and 2008; Turquoise, was launched on 22 September 2008 by a consortium of investment banks and then acquired by the LSE in late 2009; and Nasdaq OMX Europe, which belonged to the Nasdaq OMX group, was launched on 1 October 2008 and then closed on 21 May 2010. In less than two years, between the end of 2007 and the beginning of 2009, European stock markets rapidly shifted from quasi-consolidated markets to fragmented markets. In that, the implementation of MiFID is a unique event of shifting from consolidated markets to fragmented markets within a relatively short

³ For that reason, they are sometimes designated in the press as the “Next 80”.

period of time, and this specific feature makes it an ideal ground for original research on fragmentation.

While mainly related to the rise of AT, the shift to fragmented markets was legally allowed by the enforcement of MiFID on 1 November 2007, which created a competitive environment for trading systems and services. More precisely, MiFID's enforcement has changed the European trading industry in three key ways:⁴

- (1) it allowed free competition between exchanges by abolishing the concentration rule⁵ which forced any regulated trade to be executed on the primary exchange;
- (2) it offered a regulatory framework for internalization;
- (3) it extended post-trade transparency duties to OTC trades in regulated stocks and allowed entities other than primary exchanges to report trades.

To implement key points (1) and (2), MiFID has recognized three types of order execution venues in Europe: Regulated Markets (RMs), Multilateral Trading Facilities (MTFs), and Systematic Internalizers (SIs).⁶ Any trade executed in a RM, on a MTF, or with a SI falls under regulation while trades executed outside those venues are OTC.

⁴ For details and comparison with RegNMS, the US counterpart of MiFID, refer to Davies (2008) and Petrella (2010).

⁵ When a concentration rule is in place, all trades have to be executed in the order book of the primary exchange, even if they were previously negotiated off the exchange. Prior to MiFID, some continental European countries, such as France, Italy, and Spain, had a concentration rule in application of a provision in the 1993 Investment Services Directive (ISD). This provision permitted (but did not mandate) individual member states of the European Economic Area to require orders from investors in that member state to be executed only on regulated markets. As in most countries, the only regulated market was the primary exchange, the implementation of the provision implied that trading was concentrated on the primary exchange and that competition from alternative platforms was impossible. Such concentration rules disappeared with MiFID to allow competition.

⁶ The future revised version of MiFID, known as MiFID II and due to entry into application on 3rd January 2018, will also recognize Organized Trading Facilities (OTFs) for bonds, structured finance products, emission allowances or derivatives. The main distinction between RMs and MTFs on the one hand and OTFs on the other is that the execution of orders on an OTF is carried out on a discretionary basis. For instance, the operator of an OTF that crosses clients' orders may decide if, when, and how much of two or more orders it wants to match within its system.

RMs and MTFs are multilateral trading systems with similar functionalities. Both RMs and MTFs are allowed to organize primary listings. However, they differ in that RMs are legally authorized to list regulated financial instruments while financial products listed by MTFs would be considered as unregulated instruments. In practice, only RMs offer primary listing services. MTFs prefer not to do so and they may be viewed as the equivalent of ECNs in the U.S.

SIs are investment firms which, on an organized, frequent and systematic basis, execute client orders outside RMs or MTFs, either on their principal accounts or against other clients' orders. Legally, a SI does not have to be designated by a RM, and an institution can be a SI for securities listed on different stock exchanges. Creating the legal status of SIs has institutionalized internalization. In counterpart, MiFID treats SIs as mini-exchanges and imposes pre-trade and post-trade transparency requirements on them.⁷

In addition, under the post-trade transparency rules introduced by MiFID, all transactions in regulated financial instruments must be reported, even if they are carried out OTC. Such disclosures do not have to be made with the regulated primary market exclusively; they may be made by using proprietary resources or the services of a MiFID-compliant trade reporting facility (TRF). This legislation on transaction reporting led to the creation of several TRFs that offer MiFID-compliant services. The largest of these TRFs by market share, at the time of the study, was the BOAT platform of Markit, a financial information provider.⁸

3.3. Market design of primary exchanges

Primary exchanges considered in this study, that is the LSE for FTSE 100 securities and Euronext for CAC 40 and SBF 80 securities, are RMs under MiFID. They both run electronic

⁷ The same pre- and post-trade transparency requirements apply to all venues except dark pools. Dark pools are MTFs which trading mechanisms do not generate price discovery and they are exempt from pre-trade transparency rules.

⁸ Markit decided to close BOAT in late 2013. The largest trade reporting service in Europe is now offered by BATS Chi-X.

order books on which buy and sell orders are continuously matched from the open to the close according to the price/time priority rules. Trading sessions commence and finish with call auctions. On both the LSE and Euronext, off-order-book trading is permitted but with different rules and practices. At the LSE, an important dealership activity has always existed simultaneous to order book trading since its introduction in 1997 (cf. Gajewski and Gresse, 2007). Broker-dealers execute a significant part of the order flow outside the book and the majority of retail orders are routed to dealers specialized in retail trading, the so-called Retail Service Providers. Off-order-book trading is less developed on Euronext, where the order book market model has prevailed for a long time, and where concentration rules existed before the implementation of MiFID. Only two categories of trades are executed outside the Euronext central order book: block trades and VWAP trades.

3.4. Market design of other lit trading venues

MTFs operating at the time of the study were Chi-X, BATS Europe, Turquoise, and Nasdaq OMX Europe.⁹ They all run order-driven matching engines in which anonymous orders are matched continuously in time-price priority. No call auctions are organized either at the open or at the close. Their business model is based on a make/take fee structure that remunerates liquidity-providing orders and charges aggressive orders.

In addition, I also consider the quote and trade activity on the lit platforms of PLUS-Markets and Deutsche Boerse. At the time of the study, PLUS-Market was a small London-based exchange known as PLUS.¹⁰ It offered a quote-driven electronic execution venue for securities listed elsewhere in London and in continental Europe, in addition of those primarily listed on PLUS. It operated as a RM for PLUS-listed stocks, which are not part of my sample, and as a

⁹ At the current date, the most active MTFs are Chi-X, BATS Europe, and Turquoise. Nasdaq OMX Europe closed on 21 May 2010.

¹⁰ PLUS was acquired by ICAP in 2012 and rebranded as Icap Securities and Derivatives Exchange (ISDX).

MTF for European securities including those of my sample. Deutsche Boerse, the main German RM, runs an electronic order-driven trading system, Xetra, which features are very similar to those of the Euronext trading system. On Xetra, trading starts with an opening auction at 9:00am pm, then switches to continuous trading until 5:30pm, before the market closes with a closing auction.

3.5. Dark trading

Dark trades considered in this study are trades negotiated outside RMs and MTFs and executed either OTC or with a SI. OTC trades may result from a bilateral negotiating process or from order matching in an unregulated dark pool. Since MiFID implementation, those trades are reported to TRFs such as BOAT.

3.6. Market structure overview

Diagram 1 summarizes the general structure and the taxonomy of the aforementioned European stock markets.

Diagram 1 around here

4. Sample selection and data

My empirical work consists of two parts: a pre-/post-MiFID comparison and a post-MiFID time series analysis. Both parts are based on trade and quote data generously provided by Intelligent Financial Systems (IFS) for FTSE 100, CAC 40, and SBF 80 stocks.

4.1. Sample selection

In the pre-/post-MiFID comparison, the pre-MiFID period of October 2007 is compared with three post-MiFID one-month periods: January, June, and September 2009. The post-MiFID time series analysis covers 63 trading days from 1 September to 30 November 2009. Stocks which

were not continually part of their index from 2007 to 2009 were dropped from the sample. Financial stocks were excluded as very specific factors drove their liquidity and volatility during the observation periods. This selection procedure resulted in a pre/post-MiFID-comparison sample of 140 stocks of which 51 pertained to the FTSE 100, 32 to the CAC 40, and 57 to the SBF 80, and in a post-MiFID time-series-analysis sample of 152 stocks distributed between 64 FTSE-100 components, 32 CAC-40 components, and 56 SBF-80 components.

What makes the pre-/post-MiFID analysis somehow delicate is the overlapping of the 2008 financial crisis with the immediate post-MiFID period. While the best methodological practice would be to choose a post-event period immediately after the implementation of MiFID, in this particular situation it would undoubtedly bias the results by the dominant impact of the 2008 subprime crisis on liquidity, an impact that seems relatively impossible to isolate or control for. Indeed, in that period, stock markets were characterized by extreme illiquidity, independent of regulation or market architecture changes. To avoid this background effect, the post-MiFID observation periods were deliberately chosen in 2009. Choosing post-MiFID periods in 2009 has however the disadvantage of introducing a potential time trend bias in the comparison of spreads. This is overcome in two ways: (1) by a difference-in-differences test with a control sample unaffected by fragmentation in 2009; (2) by a multivariate test which controls for the determinants of liquidity and thus captures long-term time effects through those factors.

The pre-MiFID period of October 2007 is characterized by a negligible level of fragmentation: it comes just after the start-up of Chi-X but precedes the launch of other MTFs, namely BATS Europe, Nasdaq OMX Europe, and Turquoise. The three post-MiFID monthly periods come after the launch of all MTFs considered in the study and are characterized by

increasing levels of fragmentation. Diagram 2 presents the chronology of the observation periods, the alternative platform inceptions, and MiFID implementation.

Diagram 2 around here

4.2. Daily data, high-frequency data, and data filtering

Daily closing prices, adjustment factors for corporate actions, and market values were collected from Datastream. The high-frequency data provided by IFS were generated from the original data flows of Euronext, the LSE, Deutsche Boerse, Chi-X, Turquoise, Nasdaq OMX Europe, BATS Europe, and PLUS. These data include transaction and best-limit data time-stamped to the second. In addition, the IFS data comprise trades reported to BOAT. For all trading venues, batch auction trades and overnight transactions have been eliminated from the dataset. Only continuous trades are further considered for the computation of average effective spreads, average trade size, and fragmentation variables used in the econometric analysis. A more detailed description of the data and their filtering can be found in Part 1 of the internet appendix.

5. Fragmentation and liquidity measures

I now present the metrics used to measure lit and dark fragmentation. I then describe the way traditional measures of liquidity are adapted to measure liquidity in a multi-market environment.

5.1. Measuring market fragmentation

Lit fragmentation is measured by using the reciprocal of a Herfindhal-Hirschman index based on RMs and MTFs market shares in the lit order flow. This fragmentation index is calculated as one divided by the sum of the squared market shares of all RMs and MTFs covered by the database. The formula of this index writes as follows:

$$LitFrag = 1 / \sum_k \left(\frac{V_k}{\sum_j V_j} \right)^2 \quad (1)$$

where V_k and V_j denotes the volumes traded on markets k and j respectively, $\sum_j V_j$ represents the

total volume traded in all markets under consideration, and $\frac{V_k}{\sum_j V_j}$ is the market share of market k

among those markets. Over the four monthly periods used for the pre/post-MiFID comparison, levels of market fragmentation by stock index are calculated as cross-sample capitalization-weighted means of individual stocks' fragmentation indices.

In the post-MiFID observation periods, the weight of dark fragmentation is estimated by using BOAT trade report data and LSE trade-reporting-service data.¹¹ The traded quantities reported by those reporting services were aggregated and reported to total trading volumes to calculate the share of dark trading, denoted *Dark*. Measure *Dark* covers various types of dark trading such as trades executed by SIs, traditional OTC trades,¹² but also unregulated dark pool trades.

5.2. Liquidity measures

Liquidity is measured for two categories of traders: local traders who connect to the primary exchange only, and cross-market traders who are connected to several trading venues. Three metrics of liquidity are considered: quoted spreads, effective spreads, and depth displayed at the

¹¹ BOAT and the LSE reporting services do not account for all dark volumes. Data from dark pools such as POSIT, Liquidnet, or Chi-X Delta, are not covered, but the total market share of dark MTFs in trading volumes did not exceed a few percent at the time of the study, so that not holding their data should not bias the results. Furthermore, BOAT was the market leader in post-trade reporting at the time of the study so that the estimate can be considered as being in the right order of magnitude.

¹² SI trades and OTC trades are similar on two aspects: negotiation is bilateral in both cases and none of those trades get cleared.

best quotes. These metrics are determined locally in each primary market in the perspective of local traders and across markets by optimizing the prices of all competing markets.

5.2.1. Global and local quoted spreads

In the data provided by IFS, the quoted spreads of each market are observable at every second. The spread observed on the primary market will be referred to as the local spread and denoted *LQS*. Each second, the highest bid price and the lowest ask price across all competing markets are determined. The difference between these two prices divided by their mid value is the quoted spread resulting from matching quotes on all markets. It will be referred to as the global or consolidated quoted spread and denoted *GQS* in the remainder of the study, as contrasted with the local quoted spread. In the pre/post-MiFID comparison, global and local quoted spreads are averaged per stock and per month. Monthly average quoted spreads are then computed for each stock index by weighting the averages of the individual stocks by market capitalization.¹³ In the post-MiFID time-series analysis, local and global quoted spreads are averaged per stock and per day.

5.2.2. Global and local effective spreads

The effective spread of a given transaction is twice the difference between the transaction price and the mid quote prevailing at the time of the transaction, measured as a percentage of this median price. The mid quote used as the baseline is a cross-market consolidated mid quote, i.e., the mid-point of the best bid and ask prices, all markets combined. Effective spreads are averaged in the same way as quoted spreads, except that they are weighted by transaction size. The transaction universe used to calculate those averages is reduced to trades executed on-book in the continuous session. OTC, internalized, and off-book transactions are excluded. Local

¹³ Market capitalization for weighting purposes is the average of the stock's market capitalization on 1 October 2007 and its market capitalization on 30 September 2009.

effective spreads, denoted *LES*, are obtained by averaging the effective spreads of the transactions executed on a given primary exchange, while global effective spreads (*GES*) are obtained by averaging the spreads of the trades from all markets.

5.2.3. Global and local depth at best limits

Depth is measured by the sum of the visible quantities associated with the best bid and ask prices. Hidden quantities and quantities displayed at other prices deeper in the book are not available in my dataset. Therefore, term “depth” will always refer to best-limit depth in the remainder of the paper, and conclusions drawn on depth have to be considered as valid for top-of-the-book depth only.

To ensure that depth is comparable between stocks regardless of price level, it is expressed in terms of capital, specifically in thousands of euros, by multiplying the bid quantity by the bid price and the ask quantity by the ask price. Local depth (*LD*) is computed in the traditional way by considering the bid and offer quantities of the primary exchange. Global depth (also denominated consolidated depth and denoted *GD*) is determined by aggregating the quantities demanded at the best bid limit on all markets quoting the best bid price and the quantities offered at the best ask limit on all markets quoting the best ask price. Average depths are then calculated using the same procedure as for average quoted spreads.

6. Shifting from monopolistic-market trading to multi-market trading and contemporaneous changes in liquidity

Table 1 lists the market shares in lit trading of all markets and provides, in its third column, a fragmentation index over the four monthly observation periods for the three stock samples. Those figures confirm that the global market structure rapidly shifted from quasi-consolidated primary markets to fragmented markets between 2007 and 2009. Fragmentation was substantial

in June and September 2009 (around 1.9 for the CAC 40 and the FTSE 100 and between 1.4 and 1.5 for the SBF 80 securities) while it was still at an intermediate level in January 2009. The primary market's share in the lit volumes of the two large-capitalization samples fell from over 95% in October 2007 to around 70% in the final period. For French mid-capitalization equities, the primary market's share remained above 80% in September 2009. The share of volumes executed on MTFs stabilized at over 25% for CAC 40 constituents and reached about 30% for FTSE 100 constituents, but did not exceed 18% for the SBF 80 stocks.

Table 1 about here

In Table 1, the LSE market share is separated into volumes executed on the LSE order book (called SETS) and volumes executed by LSE members outside the book. This breakdown shows that the structure of the order flow on the LSE changed radically between 2007 and 2009. In October 2007, over 40% of trading volumes were executed by LSE members off the order book, whereas by September 2009, no more than 13.6% of the LSE 69% market share were traded off-book. According to these figures, the SETS system did not really lose market share, but the off-book order flow diminished as the competition from other trading venues gained ground, resulting in increased automation of securities trading.

Table 2 reports the market shares of primary markets, MTFs, and dark trading, in September 2009, for FTSE 100, CAC 40, and SBF 80 stocks. Part A presents the market shares in the total trading volumes including the opening and closing auctions of the primary market while Part B presents the market shares in continuous trading only. The primary market is found to have a majority share of order-flow execution in all the samples. When calculating a fragmentation index for the total order flow including dark trading,¹⁴ the fragmentation levels of FTSE 100

¹⁴ This fragmentation index is computed considering that all dark trades are executed in an additional trading venue. The market share of dark trading in the total traded volume is thus added at the denominator of the index.

stocks (index of 2.7) and CAC 40 stocks (index of 2.65) are comparable. In contrast, the SBF 80 middle capitalization stocks are much less fragmented (index of 1.89). Euronext was still capturing over 70% of all volumes for those stocks in September 2009, while its market share had decreased to 55% for CAC 40 stocks.

Table 2 about here

The main part of lit fragmentation is caused by no more than three MTFs: Chi-X, Turquoise, and BATS Europe. There is virtually no direct order flow competition between RMs. Euronext loses a few trades to the LSE and Deutsche Boerse, but no more than 1-2% of volume, and Euronext and Deutsche Boerse have no market share in LSE stocks. Dark trades account for around 20% of total volumes – including auction, pre-open, and post-close trades – on the CAC 40 and FTSE 100. The share is slightly lower than 20% for UK large stocks and significantly higher (approximately 24%) for French large stocks.

6.1. Changes in liquidity: Univariate tests

Despite a sharp decline in trading volumes between October 2007 and September 2009 (–59% for FTSE 100 stocks, –28% for CAC 40 stocks and –32% for SBF 80 stocks, cf. Table 2), global spreads as well as the local spreads of primary markets, reported in Table 3, narrowed between October 2007 and September 2009 for the three samples. The biggest decline was on the FTSE 100, for which the average global quoted spread fell from 9.21 to 5.43 bps and the average local quoted spread fell from 9.21 to 7.07 bps. Effective spreads contracted by less than quoted spreads but nonetheless narrowed significantly, except in the case of SBF 80 medium capitalization stocks for which they remained unchanged. The decline in spreads is especially noteworthy given that the average volatility of individual stocks was higher in September 2009 than in October 2007.

Table 3 about here

Furthermore, as a robustness check, I also test the change in spreads through a difference-in-differences approach by using a control sample of Spanish stocks. The non-financial stocks pertaining to the IBEX 35 and IBEX Medium Cap indices¹⁵ from October 2007 to September 2009, that is 24 securities in total, have been selected as control stocks because, although MiFID applies to them, their order flow remained fully concentrated on the Bolsas y Mercados Españoles (BME)¹⁶ in the years following the implementation of MiFID. The same liquidity measures as for the FTSE 100, CAC 40, and SBF 80 stocks are calculated for the control sample. Global measures are taken as equal to the local measures on the primary exchange as it is the only active market for those stocks. *t*-tests are then conducted on the differences between mean liquidity variations for the tested samples and mean liquidity variations for the control sample. Results are reported in Table 3. They not only confirm our first findings but strongly reinforce them: in most cases, the statistical significance is greater with the difference-in-differences test than with the simple difference test, and the effective spreads of SBF 80 stocks turn out to be significantly decreasing in the fragmented September 2009 period relative to control stocks.

Spreads declined steadily over the three months observed in 2009, despite high volatility in January and June 2009. Spreads did widen for Euronext stocks in January 2009, but this was weak compared with the surge in volatility over that period, and with the difference-in-differences approach, this effect vanishes. In comparison with the spreads of the control stocks, those of the FTSE 100 and CAC 40 stocks significantly declined in January and June 2009

¹⁵ IBEX 35 and IBEX Medium Cap are the main stock indices of the Madrid stock exchange known under the brand name of BME. The IBEX 35 index is composed of the 35 largest Spanish stocks while the next 20 largest Spanish stocks compose the IBEX Medium Cap index. Both are capitalization-weighted.

¹⁶ BME kindly provided best bid and ask quote and tick-by-tick trade data for the control sample over the four observation periods of October 2007, January 2009, June 2009, and September 2009.

already, and those of the SBF 80 started declining in June 2009 when their fragmentation became notable.

At first sight, depth did not display such a favorable change over the same period. Between October 2007 and September 2009, as shown in Table 3, average consolidated depth fell by 73.0% for FTSE 100 stocks, by 55.2% for CAC 40 stocks, and by 39.7% for the SBF 80 stocks. Local depth showed a similar change between the two periods. However, the difference-in-differences approach brings about a completely opposite view: the depth of tested stocks decreased less than that of non-fragmented control stocks between 2007 and 2009, this difference being statistically significant for Euronext stocks.

6.2. Changes in liquidity: A multivariate simple test

As a complement to univariate tests, I run fixed-effects panel regressions of monthly average measures of spreads and depth. Given that post-MiFID periods do not immediately follow the pre-MiFID period, this simple multivariate test is essential to check that changes in liquidity factors, such as volume, volatility, and price level, are not driving the univariate test's findings. Those factors are controlled for in the following model:

$$Liq_{im} = \alpha_i + \alpha_1 \sigma_{im} + \alpha_2 \ln V_{im} + \alpha_3 1/P_{im} + \alpha_{01/09} B_{Jan09} + \alpha_{06/09} B_{Jun09} + \alpha_{09/09} B_{Sep09} + \varepsilon_{im}, \quad (2)$$

where Liq_{im} is alternatively the average global quoted spread (GQS_{im}), the average global effective spread (GES_{im}), the average global depth (GD_{im}), the average local quoted spread (LQS_{im}), the average local effective spread (LES_{im}), and the average local depth (LD_{im}), for stock i over month m ; σ_{im} measures volatility by the standard deviation of logarithmic daily closing returns for stock i over month m ; $\ln V_{im}$ is the logarithm of the total trading volume in

euros for stock i over month m ; $1/P_{im}$ is the reciprocal of the average primary market's closing price of stock i during month m . B_{Jan09} , B_{Jun09} , and B_{Sep09} are binary variables equal to one in January 2009, June 2009, and September 2009 respectively, and set to zero otherwise. α_1 , α_2 , and α_3 are the coefficients of the control variables. $\alpha_{01/09}$, $\alpha_{01/09}$, and $\alpha_{01/09}$ denote the respective coefficients of B_{Jan09} , B_{Jun09} , and B_{Sep09} . The panel regressions are run with a fixed effect per stock (α_i) and with data covering the four month of October 2007, January 2009, June 2009, and September 2009. Dummies B_{Jan09} , B_{Jun09} , and B_{Sep09} play the role of fixed effects per month.

The estimates of the regressions of consolidated liquidity measures for the pooled sample and by stock index are reported in Panel A of Table 4. They indicate that global spreads decreased in the post-MiFID period with an increasing statistical and economic significance from January 2009 to September 2009. This spread improvement is most significant for FTSE 100 components. In contrast, SBF 80 mid-cap stocks do not exhibit significant spread reduction except for their global quoted spreads in September 2009. Global depth decreased significantly but this decrease did not follow the same pattern across time as the decrease in spreads: (1) the statistical significance of the decrease in depth was already as high in January 2009 as in later periods; (2) the economic significance of this depth reduction declined from January to September 2009. Another difference with spreads lies in that the decrease in depth is highly significant for the three stock indices and not for large-capitalization indices only.

Those observations, in particular the non-synchronization of changes in spreads with changes in depth, lead me to hypothesize that those changes are not driven by the same factors. Spread reductions seem to relate to the level of market competition as this market competition was

weaker for the SBF 80 stocks in all periods and continually increased from January to September 2009 for all indices. At the opposite, most of the fall in depth happened before January 2009 for all indices and its statistical significance is not weaker for French medium-capitalization stocks, suggesting that the reduction in depth most probably has other determinants than the reduction in spreads, those determinants remaining to be determined at this stage. This conjecture is consistent with the difference-in-differences finding that the depth of sampled stocks decreased less than that of control stocks after MiFID enforcement.

Table 4 about here

Panel B of Table 4 reports the results for the regressions of local liquidity. These regressions produce the same findings with the difference that the coefficients of the monthly dummies have a slightly lower significance.

7. Comparing the pre-MiFID monopolistic-market liquidity with the post-MiFID multi-market liquidity: A two-stage panel regression analysis

In order to check to what extent the decrease in spreads and depth observed between October 2007 and September 2009 is actually assignable to market fragmentation, I complete the test of Section 6 with a two-stage regression model that accounts for the co-determination of fragmentation and liquidity. As widely discussed by O'Hara and Ye (2011, p.462), "An immediate challenge to testing for fragmentation effects on market quality are endogeneity issues". When testing the impact of market fragmentation on liquidity, endogeneity may arise at several levels: (1) at the firm level, if for example, the trading in large stocks is more fragmented, finding greater liquidity for more fragmented stocks could simply be the outcome of large stocks being more liquid; (2) at the order level, the choice of routing an order to a given venue or of splitting it between platforms is endogenous to the relative level of liquidity at each trading

venue; (3) at the market level, liquidity and fragmentation may be co-determined, with not only fragmentation impacting liquidity but also liquidity determining fragmentation. For instance, insufficient liquidity in the primary market may induce traders to go to other markets. Conversely, if newly-established trading systems behave as satellite markets and therefore need a critical mass of orders to attract order flow, higher liquidity may be required to increase fragmentation. In brief, if the level of fragmentation is partly endogenous to liquidity, the residual terms of regressions of liquidity measures onto fragmentation measures may correlate with the independent-considered fragmentation variable. In order to remedy this potential bias, I implement a two-stage regression model in which: (1) the fragmentation index is first regressed on instrumental variables; and (2) the predicted values of fragmentation obtained from this first-stage regression serve as the regressor in second-stage regressions of liquidity measures. Those regressions, conducted on monthly averages per stock over the four months of October 2007, January 2009, June 2009, and September 2009, are designed in the same spirit as the regressions of Sub-section 6.2, with the intention of explaining the macro-trend in liquidity from the pre-MiFID period to the post-MiFID periods.

7.2. First-stage regressions

In the first stage, the fragmentation index of the lit order flow for stock i in month m ,¹⁷ denoted $LitFrag_{im}$, is modeled as follows:

$$LitFrag_{im} = a_0 + a_1 \ln MV_i + a_2 \ln V_{im} + a_3 \ln TS_{im} + a_4 Tick_{im} + a_5 MN_{im} + a_6 MC_{im} + \eta_{im}^{frag}, \quad (3)$$

where $\ln MV_i$ is the logarithm of stock i 's market value measured as the median between the start-of-month euro market capitalization in October 2007 and the end-of-month euro market

¹⁷ This fragmentation index is the reciprocal of a Herfindhal index based on RMs' and MTFs' market shares as defined in Sub-section 5.1.

capitalization in September 2009; $\ln V_{im}$ denotes the logarithmic monthly traded volume; $\ln TS_{im}$ denotes the logarithmic average trade size;¹⁸ MN_{im} is the average number of trading platforms on which traders could find quotes for stock i during month m ; and MC_{im} measures the intensity of the competition between those platforms as the average number of markets quoting either the best bid price or the best ask price divided by the average number of markets quoting stock i ; $Tick_{im}$ is the average tick size relative to price on the primary exchange; and η_{im}^{frag} is the error term. Several minimum-tick-size changes were implemented by Euronext and the LSE in 2008 and 2009. As a result, $Tick_{im}$ varies across both cross-sections and monthly periods. The complete history of those tick changes is provided in Part 2 of the internet appendix. The prediction from first-stage regression (3), denoted $Lit\hat{Frag}_{im}$, then serves as a regressor for liquidity measures in the second stage.

In this first stage, I also model a proxy for AT. AT has been proved to impact liquidity (Hendershott et al., 2011). Therefore, AT and fragmentation may have combined or competing effects on liquidity. I control for this by including an AT measure in the second-stage regressions of liquidity measures. To this aim, I proxy the level of AT by the negative of the turnover – euro volume in percentage of market value – of stock i in month m divided by the total number of quote changes per minute on both the primary exchange and Chi-X. This proxy, denoted AT_{im} , is built in the spirit of the “*algo_trade_{it}*” measure of Hendershott et al. (2011), defined as the negative of dollar volume per electronic message, measured in hundreds of dollars. In the

¹⁸ In Regression (3), $\ln V_{im}$ and $\ln TS_{im}$ are the residuals of preliminary OLS regressions of respectively the average daily volume in logarithm and the average euro trade size in logarithm on $\ln MV_i$. This procedure was designed to eliminate colinearity. $\ln V_{im}$ and $\ln TS_{im}$ should be understood as the components of the logarithmic traded volume and the logarithmic trade size which are not correlated with market value.

absence of data on electronic messages, I use the frequency of quote changes¹⁹ in lieu of the number of messages at the denominator of the variable. To avoid a dominant effect of volume in the variance of the proxy, I scale volume by market value at the numerator.

To address any potential endogeneity, AT_{im} is modeled in a first stage regression with the same instrumental variables as fragmentation (Equation 3), except volume as my AT proxy is already scaled by volume. The same instruments can serve for AT and lit fragmentation because lit fragmentation is highly determined by AT, as suggested by Menkveld (2013). Indeed, algorithmic or high-frequency traders are the most eager for the low latency offered by alternative trading systems and the most likely to slice and dice orders across markets or to supply liquidity at several venues. The value of AT_{im} predicted from this first-stage regression and denoted $\hat{AT}_{im}$ is then used as a control variable in second-stage regressions.

7.3. Choice of instrumental variables

The rationale behind choosing the instruments of Equation (3) for lit fragmentation and AT is the following.

First, the effective entry in business of new electronic order books, measured by MN_{im} , and their ability to be competitive in quotes relative to incumbent markets, measured by MC_{im} , are natural instruments for both AT and lit fragmentation. New electronic order books use technologies tailored for AT, and by attracting AT, they contribute to fragment the order flow. I consider them as exogenous factors because they are greatly driven by capital investments, marketing efforts, and corporate strategies implemented by the trading industry in the background of markets.

¹⁹ A quote change is considered to occur any time a price or a quantity is modified at one of the best quotes either on the primary exchange or on Chi-X.

Second, market size ($\ln MV_i$) is an inevitable instrument for AT. Indeed, the portion of AT corresponding to high-frequency market making is expected to be more profitable and thus more developed for heavily traded stocks. Consistently, Brogaard et al. (2014) observe that high-frequency traders are more concentrated in large stocks. Market size and trading volume ($\ln V_{im}$) are also commonly used as regressors for fragmentation (cf. Madhavan, 2012; O'Hara and Ye, 2011) as the order flow of larger and more traded stocks usually fragment more.

Third, trade size ($\ln TS_{im}$) is a direct indicator of the presence of high-frequency traders because they trade in small lots by nature. For that reason, it serves as an instrument for both AT and lit fragmentation. Order size has also often been considered as a determinant of routing strategies, leading several researchers (Madhavan and Cheng, 1997; Bessembinder, 2003; O'Hara and Ye, 2011) to use trade size as a first-stage regression instrument in order to address the order-level endogeneity issue.

The last instrument considered is the average relative tick size ($Tick_{im}$). According to Yao and Ye (2014), tick size is a key factor in AT and multi-platform market-making strategies. O'Hara et al. (2015) observe that high-frequency trading firms that operate as market makers on the NYSE take on a more prominent role in liquidity provision for stocks with larger relative tick sizes. Peng, Jarnećić, and Liu (2015) find that tick-constrained stocks tend to trade more on Chi-X to avoid queuing, proving that tick size impacts fragmentation. This is confirmed by O'Hara et al. (2015) who find that, with a larger relative tick size, some volume shifts from the primary market to other trading venues. This first-stage instrument is of particular interest as several tick size changes occurred inside my observation period (see the internet appendix).

7.4. Second-stage regressions

The second-stage regressions of liquidity measures are designed as follows for the pooled sample:

$$Liq_{im} = \beta_m + \beta_c \cdot Controls_{im} + \beta_{AT} \hat{AT}_{im} + \beta_{frag} Lit\hat{Frag}_{im} + \varphi_{im}^1. \quad (4)$$

β_m is a fixed effect per month. $Controls_{im}$ is a vector of control variables comprising: σ_{im} , the close price return volatility of stock i in month m ; market size $\ln MV_i$; traded volume $\ln V_{im}$ as defined in equation (3); and $1/P_{im}$, the reciprocal of the average closing price. β_c is a 4-dimension coefficient vector. Variable $\hat{AT}_{im}$ controls for the level of AT. $Lit\hat{Frag}_{im}$, which measures lit fragmentation, is the main variable of interest. φ_{im}^1 is a residual term.

For the analysis by index sub-samples, a variable interacting market size and fragmentation is introduced to test whether there exists differentiated affects according to size:

$$Liq_{im} = \beta_m + \beta_c \cdot Controls_{im} + \beta_{hft} \hat{AT}_{im} + \beta_{frag} Lit\hat{Frag}_{im} + \beta_{size} Size_i \times Lit\hat{Frag}_{im} + \varphi_{im}^2. \quad (5)$$

$Size_i$ denotes the market size quartile of stock i inside its index. It takes four discrete values from one to four, one corresponding to the quartile of the largest capitalizations and four corresponding to the quartile of the smallest.

7.5. Results

First-stage regressions, reported in Table 5-1, are estimated for each index separately. Most significant factors to explain fragmentation and AT are market size, the number of active markets, and tick size. The intensity of the competition in quotes between platforms is only significant for fragmentation. Coefficient signs are in the expected direction except that of tick size which indicates a decreasing relation of fragmentation and AT with tick size. This likely

reflects a dominant cross-sectional effect according to which the largest stocks, which are more fragmented and more algo-traded, also have the smallest tick sizes.

Table 5-1 about here

Second-stage regressions are estimated for global and local liquidity measures. Regression (4) is run over the total pooled sample and each index sub-sample whereas regression (5) is run on a by-index basis only. Results are reported in Table 5-2.

Table 5-2 about here

Consistent with Hendershott et al. (2011), all regressions show that spreads and depth decrease with AT. After controlling for AT, all liquidity measures either improve with lit fragmentation or are not significantly affected. The results for the pooled sample indicate that, global and local spreads significantly tighten with lit fragmentation, coefficient β_{frag} being significantly negative at the 1% level for all spread measures. The economic significance of the beneficial impact of fragmentation on spreads is however much greater than that of AT. For instance increasing the AT variable by one standard-deviation engenders a decrease in global quoted spreads of -0.74 basis points while, in comparison, increasing lit fragmentation by one standard-deviation reduces global quoted spreads by -4.63 basis points. Depth is significantly affected by lit fragmentation neither at the global and nor at the local level.

This either beneficial or not significant effect of lit fragmentation on liquidity holds in the by-index analysis, with the exception of a harmful effect on the depth of Euronext small stocks (SBF 80 index). The stocks that most benefit from fragmentation are those of the FTSE 100, for which all measures of liquidity significantly improve with fragmentation at the 1% statistical threshold. For CAC 40 stocks, spreads are not significantly impacted but their global depth increases with fragmentation. β_{size} is significantly negative (positive) in FTSE 100 spread

(depth) regressions, which indicates that liquidity benefits from lit fragmentation are more pronounced for the smallest components of the FTSE 100 index. For CAC 40 stocks, β_{size} does not significantly differ from zero in spread regressions; and it is either insignificant or significantly positive at the 10% level in depth regressions.

In contrast, comparing the findings for the FTSE 100 and CAC 40 large stocks with those for the SBF 80 medium stocks shows that lit fragmentation is more beneficial to large stocks. The global quoted spreads, the global effective spreads, and the local effective spreads of SBF 80 stocks negatively correlate to fragmentation, but converse to CAC 40 stocks, their global and local depth is adversely affected: all in all, the small stocks of Euronext benefit less from the shift to fragmented markets than the large ones. Further, the adverse effect of fragmentation on the depth of SBF 80 stocks is more pronounced for the largest components ($\beta_{size} > 0$ for global depth and $\beta_{size} > 0$ for local depth).

I then set out to investigate why the statistical significance and the economic magnitude of liquidity benefits are much greater for FTSE 100 constituents than for CAC 40 constituents. A possible explanation is that before MiFID, electronic trading was less developed at the LSE than on Euronext. According to the statistics provided in Table 2, nearly 44% of the LSE traded volume in FTSE 100 stocks were executed by dealers outside the book in October 2007. This share dramatically shrank with the expansion of alternative trading platforms to be less than 14% in September 2009. This might explain why FTSE 100 securities benefited more from lit market competition. To test this hypothesis I extended regression (5), for the FTSE 100 sub-sample, with term $\beta_{obs} Off_i \times Lit\hat{Frag}_{im}$, where Off_i is the share of off-book trading in the LSE traded volume of stock i in October 2007 and β_{obs} is the associated coefficient. Estimates are provided

in Table 5-2. β_{obs} is found to be significantly negative at the 1% statistical threshold in spread regressions while it turns out to be insignificant for depth. This supports the view that stocks with more off-book trading, that is less electronic trading before the fragmentation event, had their spreads more substantially improved by market competition.

8. A time series analysis of the relation between lit and dark fragmentation and liquidity in fragmented markets

While Sections 6 and 7 compare market liquidity in two different trading environment and regulatory regimes, this section focuses on how liquidity relates to fragmentation in a multi-market trading structure in which fragmentation has already reached a substantial level. To test this relation, I conduct panel analyses on the basis of daily fragmentation and liquidity measures over 63 trading days spreading from 1 September 2009 to 30 November 2009. The sample consists of 152 stocks distributing between 64 FTSE-100 constituents, 32 CAC-40 constituents, and 56 SBF-80 constituents, selected according to the sampling procedure described in Sub-section 4.1. For that sample and that period the data allow measuring the level of dark trading with BOAT and LSE trade report data. The analysis of this section thus examine both the effect of lit fragmentation (variable *LitFrag*) and the effect of dark fragmentation (variable *Dark*).

8.1. Regression design

To address endogeneity issues, I again implement a two-stage methodology which consists of modeling the lit and dark fragmentation daily measures as a function of instrumental variables in a first stage and using their predictions as regressors of daily liquidity measures in the second stage.

First-stage regressions are designed as follows:

$$LitFrag_{it} = c_0 + c_1 \ln MV_i + c_2 \ln V_{it} + c_3 \ln TS_{it} + c_4 MC_{it} + c_5 Tick_{it} + c_6 LitFrag_{it-1} + u_{it}^{frag}; \quad (6)$$

$$Dark_{it} = d_0 + d_1 \ln MV_i + d_2 \ln V_{it} + d_3 \ln TS_{it} + d_4 \ln DiffTS_{it} + d_5 MC_{it} + d_6 Tick_{it} + d_7 Dark_{it-1} + u_{it}^{dark}. \quad (7)$$

$LitFrag_{it}$ and $Dark_{it}$ are respectively the fragmentation index and the share of dark volumes as defined in Sub-section 5.1 for stock i on day t . V_{it} denotes volume as previously defined. MV_i equals the market capitalization of stock i on the first day of the observation period. MC_{it} is the level of competition between markets, measured by the average number of markets quoting the best bid and ask prices divided by the average number of markets with quotes. This variable is computed in the same way as in Subsection 7.2 but on a daily basis. TS_{it} is the daily average euro size of lit trades, i.e., trades executed during the continuous session on lit venues. $DiffTS_{it}$ is the difference in size between lit trades and dark trades. $Tick_{it}$ is the tick size relative to price on the primary exchange. On 1 September 2009, both Euronext and the LSE were applying a dynamic tick size schedule for the stocks of the sample (cf. the internet appendix). Therefore, $Tick_{it}$ varies across cross-sections and dates not only because it is relative to price but also because the price may switch from one price class to the other in the tick size schedule. Further, a new dynamic tick size regime was implemented by the LSE on 21 September 2009 for 12 stocks listed in the internet appendix. u_{it}^{frag} and u_{it}^{dark} are residual terms. Regressions (6) and (7) are OLS-estimated by stock index and include one lag of the dependent variable.

The reasons for using market value, volume, trade size, relative tick size, and multi-market competition level as first-stage instrumental variables are the same as those exposed in Subsection 7.3. Instrument MN , that is the average number of markets quoting prices for a stock, is abandoned here because its variance is insufficient to provide explanatory power on a daily

frequency after September 2009. The modelling of $Dark_{it}$ in Equation (7) includes an additional instrument, $DiffTS_{it}$, which measures the average trade size difference between lit and dark markets. A central motivation for trading in dark venues is to find immediacy on large trades, i.e., depth that lit venues are unable to provide. The larger dark trades relative to lit trades, the greater the ability of dark markets to provide large traders with immediacy in comparison with lit venues, and the greater the incentive for large traders to deal outside lit order books.

At the second stage, daily liquidity measures are regressed onto lit fragmentation and dark trading as predicted from the first stage and respectively denoted $LitFrag_{it}$ and $\hat{Dark}_{it}$:

$$Liq_{it} = \gamma_i + \gamma_t + \gamma_c \cdot Controls_{it} + \gamma_{AT} \hat{AT}_{it} + \gamma_{frag} \hat{LitFrag}_{it} + \gamma_{dark} \hat{Dark}_{it} + v_{it}. \quad (8)$$

Dependent variable Liq_{it} is alternatively the global or local average quoted spread, average effective spread, and average best-limit depth of stock i on day t . $Controls_{it}$ is a control variable vector composed of daily volatility measured by the mid-quote range of stock i on day t , volume calculated as the daily euro trading volume in logarithm, and price level represented by the reciprocal of the primary market's closing price of stock i on day t . AT_{it} is calculated in the same way as in Sub-section 7.2 but on a daily basis. To address endogeneity issues, the AT_{it} variable is again modeled in a first-stage regression, as a function of market size ($\ln MV_i$), trade size ($\ln TS_{it}$), relative tick size ($Tick_{it}$), and the level of competition between trading venues (MC_{it}), plus the value of AT observed the day before (AT_{it-1}). The value predicted from this first-stage procedure, denoted $\hat{AT}_{it}$, is the regressor used in (8). γ_i and γ_t denotes cross-sections and time series fixed effects respectively. v_{it} is a residual term. Standard errors are clustered by stocks. The main coefficients of interest in Regression (8) are those of variables $\hat{LitFrag}_{it}$ and $\hat{Dark}_{it}$, while $\hat{AT}_{it}$ is a control.

8.2. Results

Second-stage panel estimates²⁰ for the pooled sample and for subsamples by index are displayed in Table 6. AT is found to be associated with lower spreads but also lower depth, consistent with Hendershott et al. (2011), with the exception that the improvement in spreads is not significant for SBF 80 stocks.

Lit fragmentation is not found to deteriorate liquidity except in the case of the smallest stocks, i.e., SBF 80 stocks, for which global and local depth is negatively correlated with *LitFrag* (Panel D of Table 6). On the contrary, for the pooled sample (Panel A of Table 6), quoted spreads decrease with lit fragmentation at the 1% significance level. Effective spreads and global depth are not significantly affected. Local depth is negatively impacted but this adverse effect is related to SBF 80 stocks only.

Positive effects of lit fragmentation remain significant by sub-samples for the FTSE 100 and the CAC40. Effective spreads and depth of FTSE 100 stocks (Panel B of Table 6) improves with lit fragmentation both globally across venues and locally on the primary exchange, while their quoted spreads are not significantly affected. For CAC 40 stocks (Panel C of Table 6), lit fragmentation is beneficial to quoted spreads and global depth while it does not significantly impact effective spreads and local depth. For both groups of large stocks, the CAC 40 and the FTSE 100, it is interesting to notice that contrary to the common view, depth improves with lit fragmentation, the effect being more significant for global than for local depth. Depth reduction effects are related to AT.

As already mentioned, those positive effects do not benefit small stocks: the spreads of SBF 80 stocks (Panel D of Table 6) are not significantly affected by lit fragmentation and their depth decreases with both AT and lit fragmentation.

²⁰ First-stage estimates are not reported for sake of brevity but are available on request.

Table 6 about here

Dark fragmentation does not harm liquidity for any sample considered. It contributes to improving the liquidity of large stocks either by improving global and local depth as for FTSE 100 components, or by tightening global and local effective spreads as for CAC 40 stocks. Other measures of liquidity are not affected. Results on dark trading fail to be significant for SBF 80 components.

9. Conclusion

The coming into force of MiFID in November 2007 has caused increased competition, higher complexity, and dramatic change in trading in Europe. This study draws on high-frequency data from the most active markets for FTSE 100 constituents, CAC 40 constituents, and medium capitalization stocks of the SBF 80 index, before and after MiFID. It shows that the trading in European large equities had become substantially fragmented two years after the enforcement of MiFID. Primary exchanges, however, were still dominant players in terms of volume and price competitiveness. Three MTFs had built significant market shares: Chi-X in all categories of equities, BATS Europe more specifically in large UK securities, and Turquoise more specifically in Euronext large equities. Over the two years following MiFID, spreads decreased while depth declined dramatically. The pre/post-MiFID multivariate comparison shows that spreads narrowed gradually as the fragmentation of lit order flow increased. The spread reduction was less significant for mid-cap stocks than for large-cap stocks. Among large stocks, the decrease in spreads was more significant for LSE-listed stocks which were subject to fiercer competition from Chi-X. In contrast with spreads, the depth reduction identified with univariate tests did not increase in significance with fragmentation, meaning that it was likely driven by some other factor. This factor turns out to be AT in the rest of the analysis.

A multivariate two-stage approach involving AT and lit fragmentation measures shows that the post-MiFID reduction in spreads results more from lit fragmentation than from AT and that the post-MiFID reduction in depth results from AT. Liquidity benefits from lit fragmentation are greater for large-capitalization stocks and for stocks that had less electronic trading before MiFID. One caveat to put on about the pre/post-MiFID comparison is that it may be altered by the liquidity decline in the aftermath of the 2008 subprime crisis even though the post-MiFID periods were carefully selected. If so, the main conclusions would nevertheless be unchanged as the bias would be an underestimation of market competition benefits.

The post-MiFID time-series analysis confirms that lit market fragmentation should be viewed as value-creating competition that benefits global liquidity. In most cases, it positively impacts the local liquidity of the primary market as well. The spreads of large stocks decrease with both AT and lit fragmentation. The best-quote depth of large stocks increases with lit fragmentation, adverse effects on depth being the outcome of AT. This positive effect of fragmentation on depth is more significant for global depth across markets than for local depth on the primary exchange. Whether this inflated consolidated depth is real or artificially created by orders replicated across trading venues remains to be determined in future research.

There however are some limitations to those benefits. After an equilibrium level of market fragmentation is reached, marginal gains in liquidity become relatively low in terms of economic magnitude, even if their statistical significance is high. More importantly, lit fragmentation harms the market depth of small stocks without contributing to improving their spreads.

Finally, I find that dark fragmentation positively impacts liquidity, or at worst does not affect it. This result however leaves room for further research on dark trading and its impact on liquidity. In particular, the causal links between dark trading and the various dimensions of liquidity should be examined, and the various forms of dark trading should be distinguished.

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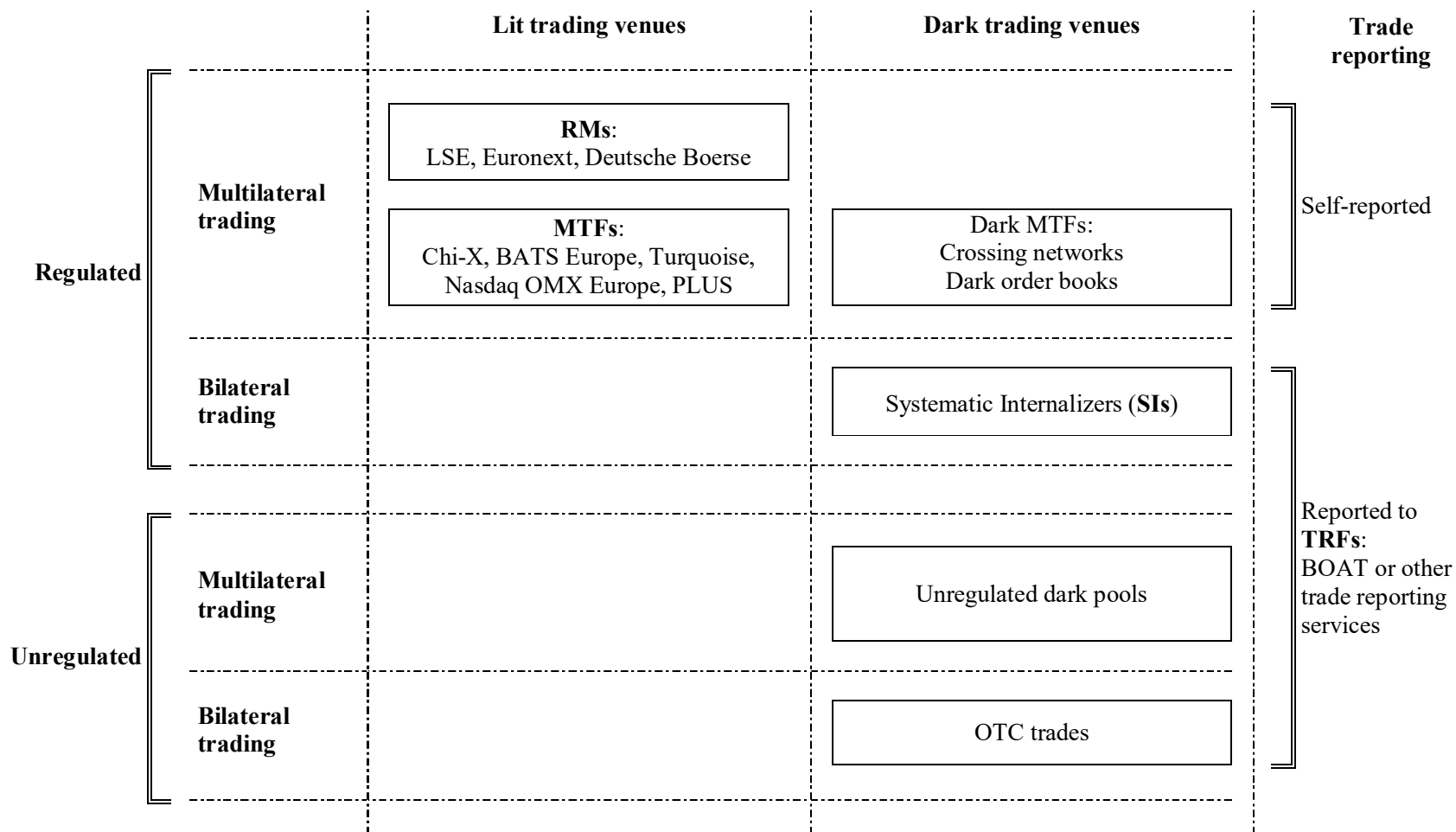
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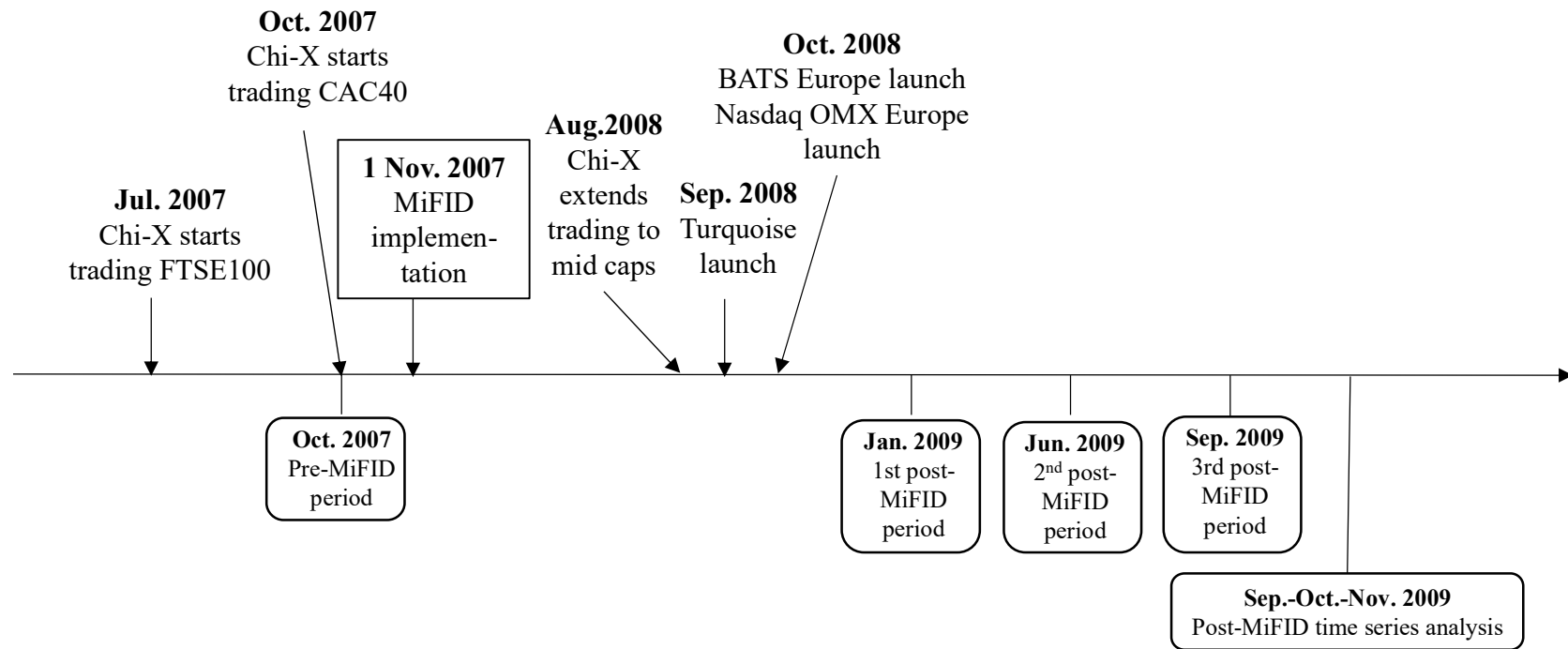
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Diagram 1. Market structure overview



Note: This diagram presents the taxonomy of European trading venues under the MiFID regulation. The markets explicitly mentioned in the diagram are those covered by the data used in the present study.

Diagram 2. Relevant dates and observation periods



Note: This timeline shows the chronology of the observation periods and the relevant events for the study (MiFID implementation and new markets' inception).

Table 1. Distribution of trading volumes between RMs and MTFs - October 2007 to September 2009

			Market shares								
	Trading volumes EUR 000	Fragmentation index				Deutsche Boerse	Chi-X	BATS Europe	Turquoise	Nasdaq OMX Eur.	PLUS
			Euronext	LSE							
FTSE 100											
Oct. 2007	191,653,000	1.00	0.00%	Total	99.99%	0.00%	0.00%	0.00%	0.00%	0.00%	0.01%
				Order book	56.46%						
				Off book	43.53%						
Jan. 2009	76,368,643	1.66	0.00%	Total	75.94%	0.00%	14.72%	1.72%	5.82%	0.29%	1.52%
				Order book	67.44%						
				Off book	8.50%						
June 2009	90,706,502	1.83	0.00%	Total	71.59%	0.00%	17.14%	4.36%	4.58%	0.71%	1.62%
				Order book	59.92%						
				Off book	11.67%						
Sep. 2009	78,769,760	1.95	0.00%	Total	68.96%	0.00%	17.26%	6.26%	5.49%	1.19%	0.84%
				Order book	55.39%						
				Off book	13.57%						
CAC 40											
Oct. 2007	105,322,607	1.07	96.45%		2.92%	0.63%	0.00%	0.00%	0.00%	0.00%	0.00%
Jan. 2009	66,111,453	1.65	76.41%		1.63%	0.61%	13.07%	1.28%	6.93%	0.08%	0.01%
June 2009	67,564,708	1.90	70.35%		1.18%	1.30%	16.42%	6.18%	3.74%	0.84%	0.00%
Sep. 2009	75,847,834	1.81	72.44%		0.72%	1.24%	15.29%	3.89%	5.21%	1.19%	0.01%
SBF 80											
Oct. 2007	17,582,701	1.00	99.99%		0.00%	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%
Jan. 2009	8,569,064	1.16	92.87%		1.15%	0.01%	5.11%	0.31%	0.51%	0.03%	0.01%
June 2009	9,625,224	1.49	80.61%		0.95%	0.02%	13.87%	2.11%	2.26%	0.19%	0.00%
Sep. 2009	11,937,753	1.44	82.47%		0.40%	0.23%	11.90%	1.30%	3.22%	0.47%	0.01%

Note: This table shows the relative market shares of trading venues for each sample of stocks in each of the observation periods. The statistics are based on a transaction universe reduced to RMs and MTFs. Transactions executed during opening and closing auctions in the primary market are included. Dark trades are excluded. The second column shows total trading volumes in thousands of euros. The third column reports fragmentation indices calculated as the reciprocal of a Herfindhal-Hirschman index based on market shares.

Table 2. Distribution of trading volumes between RMs, MTFs, and dark venues in September 2009

	Trading volumes, EUR 000	Euronext	LSE	Deutsche Boerse	Chi-X	BATS Europe	Turquoise	Nasdaq OMX Europe	PLUS	Dark trades
Part A. Auctions and continuous trading										
FTSE 100	97,610,438	0.00%	55.65%	0.00%	13.93%	5.05%	4.43%	0.96%	0.68%	19.30%
CAC 40	99,760,432	55.08%	0.55%	0.94%	11.63%	2.96%	3.96%	0.91%	0.00%	23.97%
SBF 80	13,959,377	70.53%	0.34%	0.20%	10.18%	1.11%	2.75%	0.40%	0.00%	14.48%
Part B. Continuous trading only										
FTSE 100	84,694,596	0.00%	53.35%	0.00%	16.05%	5.82%	5.10%	1.10%	0.78%	17.79%
CAC 40	83,918,213	51.51%	0.57%	1.03%	13.82%	3.52%	4.71%	1.08%	0.00%	23.75%
SBF 80	12,131,266	70.18%	0.37%	0.22%	11.71%	1.28%	3.17%	0.46%	0.01%	12.59%

Note: This table shows the relative market shares of trading venues in September 2009 for each sample of stocks. The second column shows total trading volumes in thousands of euros. All trading volumes, regulated and OTC, are included in the statistics. OTC trades and trades executed by SIs included in the statistics are those reported by BOAT and the LSE trade reporting service. They are aggregated in the “dark trades” column. Part A of the table reports market shares when primary market auction transactions are included. Part B reduces the sample to transactions executed in continuous trading.

Table 3. Liquidity comparison from October 2007 to September 2009

Sample	Period	Volatility			Quoted spreads		Effective spreads		Depth	
		Index volatility	Average stock return volatility		Global	Primary market	Global	Primary market	Global	Primary market
FTSE 100	Oct. 2007	0.91%	1.66%	Mean	0.0921%	0.0921%	0.0744%	0.0744%	531.017	531.017
	Jan. 2009	1.90%	9.99%	Mean	0.0898%	0.1129%	0.0721%	0.0716%	118.782	113.545
				Diff. from Oct. 2007	-0.0023%	0.0208%**	-0.0023%	-0.0028%	-412.235***	-417.471***
				Diff.-in-diff.	-0.0683%***	-0.0452%***	-0.0422%***	-0.0427%***	48.747	43.511
	Jun. 2009	1.16%	6.03%	Mean	0.0638%	0.0864%	0.0669%	0.0756%	137.711	159.549
				Diff. from Oct. 2007	-0.0284%***	-0.0057%	-0.0075%*	0.0012%	-393.305***	-371.468***
				Diff.-in-diff.	-0.0666%***	-0.0439%***	-0.2118%**	-0.2030%**	-109.472	-87.634
	Sep. 2009	0.77%	3.81%	Mean	0.0543%	0.0707%	0.0591%	0.0680%	142.950	134.107
				Diff. from Oct. 2007	-0.0378%***	-0.0215%***	-0.0153%***	-0.0065%	-388.067***	-396.909***
				Diff.-in-diff.	-0.037%***	-0.0209%***	-0.5847%***	-0.5758%***	173.848	165.006
CAC 40	Oct. 2007	0.70%	1.37%	Mean	0.0604%	0.0617%	0.0493%	0.0493%	132.637	137.146
	Jan. 2009	2.42%	9.74%	Mean	0.0644%	0.0943%	0.0614%	0.0612%	43.280	41.015
				Diff. from Oct. 2007	0.0040%	0.0326%***	0.0121%***	0.0119%**	-89.358***	-96.130***
				Diff.-in-diff.	-0.0619%***	-0.0333%***	-0.0277%**	-0.0279%**	371.624***	364.852**
	Jun. 2009	1.43%	4.41%	Mean	0.0502%	0.0735%	0.0506%	0.0511%	44.582	43.194
				Diff. from Oct. 2007	-0.0102%**	0.0118%**	0.0013%	0.0018%	-88.055***	-93.952***
				Diff.-in-diff.	-0.0484%***	-0.0264%***	-0.2029%**	-0.2024%**	195.778**	189.882**
	Sep. 2009	0.90%	3.58%	Mean	0.0446%	0.0595%	0.0414%	0.0408%	59.401	54.846
				Diff. from Oct. 2007	-0.0158%***	-0.0021%	-0.0079%**	-0.0085%**	-73.236***	-82.299***
				Diff.-in-diff.	-0.0152%**	-0.0016%	-0.5772%***	-0.5779%***	488.680***	479.617***

Table 3. cont'd

Sample	Period	Volatility			Quoted spreads		Effective spreads		Depth	
		Index volatility	Average stock return volatility		Global	Primary market	Global	Primary market	Global	Primary market
SBF 80	Oct. 2007	0.62%	1.63%	Mean	0.1452%	0.1454%	0.1130%	0.1129%	35.028	35.222
	Jan. 2009	1.85%	12.12%	Mean	0.2502%	0.2644%	0.2010%	0.2010%	16.937	16.919
				Diff. from Oct. 2007	0.1050%***	0.1190%***	0.0881%***	0.0881%***	-18.091***	-18.303***
				Diff.-in-diff.	0.039%**	0.0531%***	0.0482%***	0.0483%***	442.891***	442.679***
	Jun. 2009	1.37%	4.99%	Mean	0.1615%	0.1945%	0.1408%	0.1394%	17.955	18.216
				Diff. from Oct. 2007	0.0163%	0.0491%***	0.0278%*	0.0265%*	-17.073***	-17.006***
				Diff.-in-diff.	-0.0219%	0.0109%	-0.1764%*	-0.1778%*	266.761***	266.828***
	Sep. 2009	0.88%	4.85%	Mean	0.1198%	0.1399%	0.1129%	0.1117%	21.113	21.594
				Diff. from Oct. 2007	-0.0254%**	-0.0055%	0.0000%	-0.0012%	-13.915***	-13.628***
				Diff.-in-diff.	-0.0248%**	-0.0049%	-0.5694%***	-0.5705%***	548.001***	548.288***

Note: This table reports averages of fragmentation indices, volatility measures, spreads, and depth measures, by month and by sample, as well as statistics on the differences in the liquidity measures between the non-fragmented pre-MiFID period (Oct. 2007) and each fragmented post-MiFID period in 2009. The third column gives a fragmentation index calculated by taking the inverse of the sum of the squares of market shares. Volatility is measured in two ways: the fourth column shows the volatility of the index daily returns during the month; the fifth column is the capitalization-weighted average of individual stocks' daily-return volatilities. Average quoted spreads, average effective spreads, and average depths are measured globally across markets and locally in the primary market. Effective spread statistics are based on a reduced universe comprising transactions exchanged in on-book continuous trading. Depth is expressed in thousands of euros. Depth only includes disclosed quantities at the best quotes. For each month, the "Mean" line reports the means of liquidity measures; the "Diff. from Oct. 2007" line reports their mean differences between Oct. 2007 and the considered month; the "Diff.-in-diff." line reports the mean differences in differences between the tested sample and a control sample of 24 non-fragmented Spanish stocks. Cross-sectional means are weighted by capitalization. ***, **, * means that the difference in consideration is statistically different from zero according to a *t*-test at the 1%, 5%, or 10% level respectively.

Table 4. One-stage panel regressions of monthly liquidity measures around MiFID

	Global liquidity			Local liquidity		
	Quoted spread (GQS_{im})	Effective spread (GES_{im})	Best-limit depth (GD_{im})	Quoted spread (LQS_{im})	Effective spread (LES_{im})	Best-limit depth (LD_{im})
Panel A – Pooled sample						
B_{Jan09}	-0.00002 (-0.24)	0.00009 (1.11)	-0.69604*** (-14.35)	0.00015 (1.59)	0.00008 (0.99)	-0.73178*** (-16.10)
B_{Jun09}	-0.00050*** (-5.83)	-0.00013* (-1.78)	-0.61521*** (-14.60)	-0.00023*** (-2.76)	-0.00011 (-1.48)	-0.61427*** (-15.56)
B_{Sep09}	-0.00067*** (-8.58)	-0.00025*** (-3.80)	-0.50359*** (-13.22)	-0.00047*** (-6.30)	-0.00023*** (-3.44)	-0.48063*** (-13.47)
R ²	0.8353	0.8362	0.9306	0.8449	0.8333	0.9381
Panel B – FTSE 100 stocks						
B_{Jan09}	-0.00063*** (-7.38)	-0.00041*** (-6.24)	-0.79258*** (-8.05)	-0.00040*** (-4.75)	-0.00041*** (-6.19)	-0.78983*** (-9.54)
B_{Jun09}	-0.00095*** (-12.62)	-0.00049*** (8.45)	-0.60153*** (-6.90)	-0.00071*** (-9.51)	-0.00041*** (-6.95)	-0.53461*** (-7.30)
B_{Sep09}	-0.00105*** (14.15)	-0.00053*** (-9.33)	-0.54449*** (-6.33)	-0.00083*** (-11.24)	-0.00044*** (-7.54)	-0.44075*** (-6.10)
R ²	0.8772	0.8291	0.8911	0.8860	0.8056	0.9155
Panel C – CAC 40 stocks						
B_{Jan09}	-0.00008** (-2.06)	0.00001 (0.26)	-0.85670*** (-9.58)	0.00019*** (5.05)	0.00002 (0.52)	-0.92423 (-12.16)
B_{Jun09}	-0.00023*** (-7.87)	-0.00008** (-2.42)	-0.84921*** (-11.89)	-0.00001 (-0.33)	-0.00007** (-2.61)	-0.89054 (-14.67)
B_{Sep09}	-0.00023*** (-9.20)	-0.00012*** (-4.28)	-0.61695*** (-9.94)	-0.00010*** (-3.92)	-0.00014*** (-6.22)	-0.67329 (-12.75)
R ²	0.8890	0.8580	0.9017	0.9329	0.8977	0.9284

Table 4. cont'd

	Global liquidity			Local liquidity		
	Quoted spread (GQS_{im})	Effective spread (GES_{im})	Best-limit depth (GD_{im})	Quoted spread (LQS_{im})	Effective spread (LES_{im})	Best-limit depth (LD_{im})
Panel D – SBF 80 stocks						
B_{Jan09}	0.00052** (2.46)	0.00051*** (2.75)	-0.41534*** (-6.68)	0.00058*** (2.74)	0.00051*** (2.75)	-0.40662*** (-6.38)
B_{Jun09}	-0.00022 (-1.38)	0.00013 (0.91)	-0.42931*** (-9.25)	0.00010 (0.62)	0.00012 (0.88)	-0.43049 (-9.05)***
B_{Sep09}	-0.00048*** (-3.56)	-0.00005 (-0.42)	-0.31897*** (-7.95)	-0.00029*** (-2.11)	-0.00005 (-0.43)	-0.29059*** (-7.07)
R^2	0.8503	0.8466	0.8573	0.8489	0.8442	0.8541

Note: This table reports the estimates of panel regressions of global and local liquidity measures run with a fixed effect per stock and monthly dummies standing for January, June, and September 2009, respectively denoted B_{Jan09} , B_{Jun09} , and B_{Sep09} . The observation period includes those three post-MiFID months plus the pre-MiFID month of October 2007. Variables used in the regressions are monthly observations per stock for 140 non-financial equities (51 FTSE-100 components, 32 CAC-40 components, and 57 SBF-80- components). Spreads are expressed as multiples of mid prices (not in percentage). Depth is expressed in thousand euros and taken in logarithm. The set of control variables includes volatility measured by the standard deviation of closing returns, volume measured by the logarithm of the total euro trading volume; and price level measured by the reciprocal of the average primary market's closing price. ***, **, * indicates statistical significance at the 1%, 5%, or 10% level respectively. *t*-statistics are provided in brackets.

Table 5-1. Two-stage regressions of monthly liquidity measures around MiFID: First-stage estimates

Dependent variable	FTSE 100		CAC 40		SBF 80	
	Fragmentation (<i>LitFrag_{im}</i>)	Algorithmic trading (<i>AT_{im}</i>)	Fragmentation (<i>LitFrag_{im}</i>)	Algorithmic trading (<i>AT_{im}</i>)	Fragmentation (<i>LitFrag_{im}</i>)	Algorithmic trading (<i>AT_{im}</i>)
Market value ($\ln MV_i$)	-0.0072 (0.5744)	3.46E-05*** ($<.0001$)	0.0595** (0.0191)	2.61E-05** (0.0490)	0.0410*** (0.0006)	3.53E-05 (0.1354)
Traded volume ($\ln V_{im}$)	-0.0291 (0.3520)		0.0142 (0.7423)		-0.0037 (0.7579)	
Trade size ($\ln TS_{im}$)	-0.0728 (0.3450)	-8.84E-05*** (0.0011)	-0.1575 (0.2059)	1.80E-05 (0.7822)	-0.0601 (0.1861)	-4.49E-04*** ($<.0001$)
Number of active markets (MN_{im})	0.3781*** ($<.0001$)	5.38E-05*** ($<.0001$)	0.2218*** ($<.0001$)	5.81E-05*** ($<.0001$)	0.1446*** ($<.0001$)	6.42E-05*** (0.0003)
Market competition intensity (MC_{im})	1.3191*** ($<.0001$)	-4.37E-06 (0.9212)	0.6909*** (0.0002)	1.04E-04 (0.2692)	0.2904*** ($<.0001$)	-1.48E-05 (0.9011)
Tick size ($Tick_{im}$)	-269.44*** ($<.0001$)	0.0256 (0.2122)	-33.4921 (0.8030)	-0.9372*** ($<.0001$)	-96.9831*** (0.0071)	-1.2092*** ($<.0001$)
Number of observations	204	204	128	128	228	228
Adjusted R ²	84.68%	70.79%	76.29%	74.41%	76.56%	65.84%

Note: This table displays the estimates of the first-stage regressions of lit fragmentation (*LitFrag_{im}*) and algorithmic trading (*AT_{im}*) by index. Variables used in the regressions are monthly observations per stock for 140 non-financial equities (51 FTSE-100 components, 32 CAC-40 components, and 57 SBF-80 components) over the pre-MiFID month of October 2007 and the three post-MiFID months of January, June, and September 2009. *LitFrag_{im}* and *AT_{im}* are respectively the monthly fragmentation index and the monthly level of AT per stock. $\ln MV_i$ is the logarithm of stock *i*'s market value; $\ln V_{im}$ is the logarithmic monthly traded volume; $\ln TS_{im}$ is the logarithmic average trade size; MN_{im} is the average number of active trading platforms for stock *i* during month *m*; and MC_{im} is the average number of markets quoting either the best bid price or the best ask price divided by the average number of markets quoting stock *i*; $Tick_{im}$ is the average tick size relative to price on the primary exchange. ***, **, * indicates statistical significance at the 1%, 5%, or 10% level respectively. *p*-values are provided in brackets.

Table 5-2. Two-stage regressions of monthly liquidity measures around MiFID: Second-stage estimates

	Global liquidity			Local liquidity		
	Quoted spread (GQS_{im})	Effective spread (GES_{im})	Best-limit depth (GD_{im})	Quoted spread (LQS_{im})	Effective spread (LES_{im})	Best-limit depth (LD_{im})
Panel A – Pooled sample						
AT_{im}	-0.2660* (0.0540)	-0.5406*** ($<.0001$)	-3.5E+02*** ($<.0001$)	-0.2274* (0.0968)	-0.5500*** ($<.0001$)	-3.3E+02*** (0.0001)
$LitFrag_{im}$	-0.0011*** ($<.0001$)	-0.0012*** ($<.0001$)	0.0686 (0.4648)	-0.0010*** ($<.0001$)	-0.0011*** ($<.0001$)	-0.0041 (0.9648)
Adjusted R ²	67.64%	67.72%	76.99%	67.92%	66.61%	77.55%
Panel B – FTSE 100 stocks						
Regression 1						
AT_{im}	-4.2329*** (0.0044)	-3.3722*** (0.0017)	-1.5E+04*** ($<.0001$)	-3.1259** (0.0388)	-3.1377*** (0.0033)	-1.2E+04*** ($<.0001$)
$LitFrag_{im}$	-0.0005** (0.0360)	-0.0004*** (0.0076)	1.0506*** ($<.0001$)	-0.0007*** (0.0014)	-0.0005*** (0.0026)	0.3993*** (0.0071)
$Size_i \times LitFrag_{im}$	-0.0001*** (0.0095)	-0.0001** (0.0112)	-0.0221 (0.3949)	-0.0001** (0.0469)	-0.0001*** (0.0078)	0.0259 (0.2508)
Adjusted R ²	67.48%	59.68%	86.45%	67.76%	56.94%	88.77%
Regression 2						
AT_{im}	-4.5391*** (0.0015)	-3.5576*** (0.0007)	-1.5E+04*** ($<.0001$)	-3.4633** (0.0163)	-3.3106*** (0.0015)	-1.2E+04*** ($<.0001$)
$LitFrag_{im}$	0.0000 (0.8672)	-0.0001 (0.4911)	1.0702*** ($<.0001$)	-0.0002 (0.4775)	-0.0002 (0.2693)	0.4418*** (0.0094)
$Size_i \times LitFrag_{im}$	-0.0001*** (0.0014)	-0.0001*** (0.0026)	-0.0228 (0.3857)	-0.0001*** (0.0085)	-0.0001*** (0.0019)	0.0245 (0.2825)
$Off_i \times LitFrag_{im}$	-0.0009*** ($<.0001$)	-0.0005*** (0.0005)	-0.0347 (0.8363)	-0.0010*** ($<.0001$)	-0.0005*** (0.0011)	-0.0753 (0.6062)
Adjusted R ²	70.17%	61.97%	86.38%	70.88%	59.06%	88.73%

Table 5-2. cont'd

	Global liquidity			Local liquidity		
	Quoted spread ($Q\hat{Q}S_{im}$)	Effective spread (GES_{im})	Best-limit depth (GD_{im})	Quoted spread ($L\hat{Q}S_{im}$)	Effective spread (LES_{im})	Best-limit depth (LD_{im})
Panel C – CAC 40 stocks						
$\hat{AT}_{im}$	-0.5924*** ($<.0001$)	-0.6595*** ($<.0001$)	-1.8E+03*** ($<.0001$)	-0.6671*** ($<.0001$)	-0.6735*** ($<.0001$)	-1.8E+03*** ($<.0001$)
$\hat{LitFrag}_{im}$	-0.0001 (0.2522)	-0.0000 (0.6442)	0.4271** (0.0168)	-0.0001 (0.3707)	-0.0001 (0.5355)	0.1950 (0.2012)
$Size_i \times \hat{LitFrag}_{im}$	0.0000 (0.1786)	0.0000 (0.1780)	0.0149 (0.5017)	0.00003** (0.0485)	0.0000 (0.2168)	0.0336* (0.0798)
Adjusted R ²	83.34%	76.62%	89.68%	86.24%	83.20%	92.28%
Panel D – SBF 80 stocks						
$\hat{AT}_{im}$	0.0183 (0.9340)	0.02015 (0.9104)	-5.5E+02*** ($<.0001$)	0.0315 (0.8886)	0.0316 (0.8613)	-5.4E+02*** ($<.0001$)
$\hat{LitFrag}_{im}$	-0.0012** (0.0442)	-0.0011*** (0.0020)	-0.3419*** (0.0064)	-0.0010 (0.1029)	-0.0016*** (0.0017)	-0.3809*** (0.0033)
$Size_i \times \hat{LitFrag}_{im}$	0.0001 (0.7043)	0.0001 (0.6439)	0.0528** (0.0450)	0.0001 (0.6198)	0.0001 (0.6379)	0.0532* (0.0504)
Adjusted R ²	67.42%	71.31%	85.36%	66.05%	71.02%	84.76%

Note: This table displays the estimates of second-stage regressions of global (Panel A) and local (Panel B) liquidity measures on lit and dark fragmentation. Variables used in the regressions are monthly observations per stock for 140 non-financial equities (51 FTSE-100 components, 32 CAC-40 components, and 57 SBF-80 components) over the pre-MiFID month of October 2007 and the three post-MiFID months of January, June, and September 2009. Spreads are expressed as multiples of mid prices (not in percentage). Depth is expressed in thousand euros and taken in logarithm. $\hat{LitFrag}_{im}$ and $\hat{AT}_{im}$ are respectively the monthly fragmentation index and the monthly level of AT per stock as predicted from first-stage regressions. $Size_i$ is the market value quartile of stock i inside its index. Off_i is the off-book trading share in the LSE traded volume for stock i October 2007. Unreported control variables are volatility, market value, volume, and price level. ***, **, * indicates statistical significance at the 1%, 5%, or 10% level respectively. p -values are provided in brackets.

Table 6. Panel regressions of daily liquidity measures by stock index: Second-stage estimates

	Global quoted spread (GQS_{it})	Global effective spread (GES_{it})	Global best-limit depth (GD_{it})	Local quoted spread (LQS_{it})	Local effective spread (LES_{it})	Local best-limit depth (LD_{it})
Panel A – Pooled sample						
$\hat{AT}_{it}$	-0.219452** (0.0107)	-4.324404*** ($<.0001$)	-1.17E+02 (0.2278)	-0.330636 (0.0622)	-4.439523*** ($<.0001$)	-3.19E+02*** (0.0002)
$\hat{LitFrag}_{it}$	-0.000172*** ($<.0001$)	-0.000257 (0.2932)	0.029145 (0.5132)	-0.000353*** ($<.0001$)	-0.000250 (0.3237)	-0.073053** (0.0410)
$\hat{Dark}_{it}$	0.000008 (0.8778)	-0.001010*** (0.0008)	0.300548*** ($<.0001$)	-0.000062 (0.5236)	-0.001077*** (0.0005)	0.237297*** ($<.0001$)
R ²	95.59%	46.00%	99.64%	83.58%	44.47%	99.69%
Panel B – FTSE 100 stocks						
$\hat{AT}_{it}$	-1.323994* (0.0686)	-0.146456* (0.0705)	-6.38E+03*** (0.0009)	-3.131174* (0.0840)	-0.300852* (0.0941)	-5.34E+03*** (0.0004)
$\hat{LitFrag}_{it}$	-0.000075 (0.2250)	-0.000186*** (0.0005)	0.392676*** (0.0034)	-0.000026 (0.8629)	-0.000149*** (0.0076)	0.267027** (0.0107)
$\hat{Dark}_{it}$	0.000086* (0.0928)	0.000040 (0.5137)	0.373017*** (0.0004)	0.000077 (0.7282)	0.000029 (0.7068)	0.244373*** (0.0050)
R ²	96.87%	93.26%	99.69%	78.51%	91.91%	99.79%
Panel C – CAC 40 stocks						
$\hat{AT}_{it}$	-0.376678*** ($<.0001$)	-12.643322*** (0.0078)	-4.21E+02* (0.0685)	-0.427465*** ($<.0001$)	-12.889377*** (0.0072)	-1.95E+02 (0.3551)
$\hat{LitFrag}_{it}$	-0.000123*** (0.0021)	0.000368 (0.7801)	0.237864*** (0.0018)	-0.000117** (0.0103)	0.000381 (0.7725)	0.125804* (0.0693)
$\hat{Dark}_{it}$	0.000000 (0.9940)	-0.001452* (0.0672)	-0.012465 (0.8527)	0.000004 (0.8914)	-0.001518* (0.0576)	0.035580 (0.5410)
R ²	99.06%	52.91%	99.84%	99.17%	50.88%	99.86%

Table 6. cont'd

	Global quoted spread (GQS_{it})	Global effective spread (GES_{it})	Global best-limit depth (GD_{it})	Local quoted spread (LQS_{it})	Local effective spread (LES_{it})	Local best-limit depth (LD_{it})
Panel D – SBF 80 stocks						
$\hat{AT}_{it}$	-0.116682 (0.4004)	-1.005629 (0.5921)	-4.85E+02*** (0.0003)	-0.054049 (0.7305)	-0.965524 (0.6217)	-4.92E+02*** (0.0004)
$\hat{LitFrag}_{it}$	-0.000076 (0.3983)	0.000599 (0.4413)	-0.146384** (0.0320)	-0.000057 (0.5576)	0.000681 (0.4090)	-0.307074*** (<.0001)
$\hat{Dark}_{it}$	-0.000022 (0.8429)	-0.000509 (0.3231)	0.086452 (0.1990)	-0.000038 (0.7313)	-0.000552 (0.3106)	0.084050 (0.2726)
R ²	95.52%	49.33%	99.53%	95.88%	47.54%	99.47%

Note: This table reports the estimates of second-stage two-fixed-effects panel regressions of daily liquidity measures on fragmentation measures for four samples of stocks – 64 FTSE-100 stocks (Panel B), 32 CAC-40 stocks (Panel C), 56 SBF-80 stocks (Panel D), and the pooled sample made of those 152 stocks (Panel A) – over 63 trading days from 1 September to 30 November 2009. The dependent variable is alternatively the average global quoted spread, the average global effective spread, the average global depth, the average local quoted spread, the average local effective spread, and the average local depth. Spreads are expressed as multiples of mid prices (not in percentage). Depth is expressed in thousand euros and taken in logarithm. The independent fragmentation measures are the lit fragmentation index ($\hat{LitFrag}_{it}$) and the dark trading share ($\hat{Dark}_{it}$) as predicted in first-stage regressions. All regressions are controlled for price range, volume, and price level. They also control for AT as predicted in a first-stage regression ($\hat{AT}_{it}$). Standard errors are clustered by stock. ***, **, * indicates statistical significance at the 1%, 5%, or 10% level respectively. *p*-values are provided in brackets.

Internet appendix.

Part 1. Daily data, high-frequency data, and data filtering (*detailed version*)

Daily closing prices, adjustment factors for corporate actions, and market values were collected from Datastream. Prices of UK stocks are expressed in pence sterling. Some metrics used in the study require expressing all prices and volumes in euro. For that purpose, the daily foreign exchange rates of the GBP against EUR were downloaded from the Oanda website (www.oanda.com).

The high-frequency data provided by IFS were generated from the original data flows of Euronext, the LSE, Deutsche Boerse, Chi-X, Turquoise, Nasdaq OMX Europe, BATS Europe, and PLUS. The database includes transaction and best-limit data time-stamped to the second. For each stock, the data contain the trade prices and best limit quotes of all trading venues. All timestamps in the database are in UK time and hours will be expressed in UK time throughout this section. In addition, the IFS data comprise trades reported to BOAT.

Best limit data provide, at each second of the trading session from 8:00 to 16:30, the best bid price, the best ask price, and quantities associated, for every trading venue where a quote is displayed. Bid and ask prices with times that could fall inside the opening auction periods or after the end of the continuous session of the primary exchange were eliminated. As a result of this rule, I deleted from the dataset bid and ask quotes with times before 08:01 and after 16:30.

Trade data cover the same markets as quote data plus BOAT-reported trades. For each trade, the data provide the execution time, the price, the size in number of shares, as well as best bid and ask prices and displayed quantities prevailing on every sampled RM or MTF at the time of the trade. Euronext and LSE data encompass opening and closing auction transactions as well as all

continuous trades except block and VWAP trades for Euronext. LSE data include an indicator that can be used to identify trades executed on the SETS order book, off-book trades executed by LSE members, and trades reported to the LSE European Trade Reporting Service but not executed on the LSE regulated market. The third category of trades (LSE-reported trades) are of the same type as trades reported by BOAT and should be considered as OTC or internalized trades.

For all trading venues, overnight transactions, that is transactions executed before 07:00 and after 17:00, have been eliminated from the dataset. LSE-reported trades have a 2-figure trade-type code. Any LSE trade for which this code indicates that the trade results from an option exercise, a stock swap, or the cancellation of a previous trade, is deleted.

I have then identified batch auction trades. For the LSE, SETS call auction trades are flagged with a special trade code. On Euronext, batch-auction trades have to be identified with trading times. At the open, trades whose execution time is the first trade time observed at or after the official start time of the open auction period are considered as open auction trades. At the close, trades executed at the first trade time immediately following or equal to the official start time of the close auction but no later than the end time of the close auction period are identified as close auction trades. Once auction trades have been determined, other trades are classified into three groups: continuous trades executed between the open and the close auctions, pre-open trades, and post-close trades. Only continuous trades are further considered for the computation of average effective spreads, average trade size, and fragmentation variables used in the econometric analysis.

Part 2. History of minimum-tick-size changes at the London Stock Exchange (LSE) and on Euronext over the period 2007-2009

A. Tick size changes on Euronext

On 19 February 2007, Euronext started decimal trading with a fixed tick size of 0.01 for all stocks. In 2008, Euronext began the implementation of multi-decimal trading with the application of a fixed tick size of €0.005 for a limited number of stocks. This implementation took place on 28 January 2008 for some Belgian and Portuguese stocks, and on 7 April 2008 for a bunch of Dutch and French stocks. The stocks of my sample impacted by this change are listed in Table A1.

Table A.1

Stock name	ISIN code	Implementation date
ALCATEL-LUCENT	FR0000130007	7 April 2008
STMICROELECTRONICS	NL0000226223	
FRANCE TELECOM	FR0000133308	
VIVENDI	FR0000127771	
TOTAL	FR0000120271	

Source: Info-Flash Euronext 28 December 2007.

On 26 May 2008, Euronext changed the tick size of two highly liquid but lower priced stocks from €0.005 to €0.001. Those stocks are STMICROELECTRONICS in Paris and AEGON in Amsterdam (Source: Info-Flash Euronext 23 May 2008), of which the first one is my sample.

As of 1 September 2008, a fixed tick size of three decimal places was implemented for a restricted list of stocks including the components of the CAC 40 index but not those of the SBF 80. The value of this new tick size was determined depending on the average trade price over the last two months, with a value of €0.005 where the average price exceeded €10 and a value of €0.001 otherwise. This new fixed tick size of €0.005 applied to all CAC 40 securities in our sample except ALCATEL-LUCENT for which the tick size was moved to €0.001. Then, from 15 December 2008, those price-depending tick sizes were implemented on a dynamic basis for all CAC 40 and SBF 80 stocks.

Last, on 16 July 2009, Euronext applied the European harmonized tick size regime for a list of blue-chip stocks including CAC 40 and SBF 80 securities. The dynamic tick size table implemented for those stocks follows Table A.2

Table A.2

Price from	To	Tick size
€0	€9.999	€0.001
€10.000	€49.995	€0.005
€50.000	€99.99	€0.01
€10.000	∞	€0.05

Source: Info-Flash Euronext 2 July 2009.

B. Tick size changes on the LSE

Before 4 December 2007, the dynamic tick size regime presented in Table B.1 was in use for FTSE 100 securities (market segment “SET1” at the LSE).

Table B.1

From	To	Tick size
Price in GBX		
0	499.75	0.25
500.0	999.5	0.5
1,000	∞	1
Price in GBP / USD / EUR		
0	4.9975	0.0025
5.000	9.995	0.005
10.00	∞	0.01

On 4 December 2007, a new dynamic tick size schedule was introduced. Table B.2 displays the regime then implemented for market segment “SET1” to which my sample stocks belongs.

On 19 May 2008, five FTSE 100 index securities moved from the dynamic price format code schedule to specific static tick sizes as shown in Table B.3.

On 22 June 2009, 12 other FTSE 100 stocks moved from the dynamic price format code schedule to specific static tick sizes and two FTSE 100 stocks had their static tick sizes revised in compliance with Table B.4.

In July 2009, the LSE announced that it would apply the FESE tick size harmonization to FTSE 100 and FTSE 250 securities. As a consequence, from 3 August 2009, a new dynamic price format schedule complying with “FESE Table 2” (cf. Table B.5) was introduced for FTSE 100 securities. Further, a number of FTSE 100 stocks were placed on static tick sizes contained within “FESE Table 1” while others which were on a static price format codes were moved to the new dynamic price format schedule. Details of those changes are provided in Table B.6. Stocks remaining on a static tick size with no change are listed in Table B.7.

Table B.2

From	To	Tick size
Price in GBX		
0	9.99	0.01
10	199.9	0.1
200	499.75	0.25
500	999.5	0.5
1,000	∞	1
Price in GBP / USD / EUR		
0	0.0999	0.0001
0.1	4.9975	0.0025
5	9.995	0.005
10	∞	0.01

Source: LSE Service Announcement 106-07- 27 November 2007.

Table B.3

Stock name	ISIN code	Static tick size	Trading currency
BP	GB0007980591	0.25	GBX
GLAXOSMITHKLINE	GB0009252882	0.5	
HSBC HLDGS	GB0005405286	0.25	
TESCO	GB0008847096	0.1	
VODAFONE GROUP	GB00B16GWD56	0.05	

Source: LSE Service Announcement 36-08- 6 May 2008.

Table B.4

Stock name	ISIN code	Static tick size	Trading currency
ANGLO AMERICAN	GB00B1XZS820	0.5	GBX
ASTRAZENECA	GB0009895292	0.5	
BARCLAYS	GB0031348658	0.05	
BHP BILLITON	GB0000566504	0.5	
BP	GB0007980591	0.05	
BT GROUP	GB0030913577	0.05	
FRIENDS PROVIDENT ¹	GB00B3T69350	0.01	

¹ The ISIN code of Friends Provident was GB0030559776 before 15 June 2009.

HSBC HLDGS	GB0005405286	0.05
LEGAL & GENERAL	GB0005603997	0.01
LLOYDS GRP.	GB0008706128	0.01
OLD MUTUAL	GB00B77J0862	0.01
RIO TINTO	GB0007188757	0.5
ROYAL BANK SCOT	GB00B7T77214	0.005
XSTRATA	JE00B4T3BW64	0.1

Source: LSE Service Announcement 001/170609- 17 June 2009.

Table B.5

Price (in GBX/GBP/USD/EUR) from	To	Tick size
0	0.4999	0.0001
0.5	0.9995	0.0005
1	4.999	0.001
5	9.995	0.005
10	49.99	0.01
50	99.95	0.05
100	499.9	0.1
500	999.5	0.5
1,000	4,999	1
5,000	9,995	5
10,000	∞	10

Source: LSE Service Announcement 001/170709- 17 July 2009.

Table B.6

Stock name	ISIN code	New tick size	Trading currency
FRIENDS PROVIDENT	GB00B3T69350	Dynamic	GBX
LEGAL & GENERAL	GB0005603997	Dynamic	
OLD MUTUAL	GB00B77J0862	Dynamic	
ROYAL BANK SCOT	GB00B7T77214	Dynamic	
XSTRATA	JE00B4T3BW64	Dynamic	
TESCO	GB0008847096	Dynamic	
HSBC HLDGS	GB0005405286	0.1 (Static)	
ROYAL DUTCH SHELL 'A'	GB00B03MLX29	0.5 (Static)	

Source: LSE Service Announcement 001/170709- 17 July 2009

& LSE Service Announcement 001/200709- 20 July 2009

Table B.7

Stock name	ISIN code	Static tick size	Trading currency
ANGLO AMERICAN	GB00B1XZS820	0.5	GBX
ASTRAZENECA	GB0009895292	0.5	
BARCLAYS	GB0031348658	0.05	
BHP BILLITON	GB0000566504	0.5	
BP	GB0007980591	0.05	
BT GROUP	GB0030913577	0.05	
GLAXOSMITHKLINE	GB0009252882	0.5	
LLOYDS GRP.	GB0008706128	0.01	
RIO TINTO	GB0007188757	0.5	
VODAFONE GROUP	GB00B16GWD56	0.05	

Source: LSE Service Announcement 001/170709- 17 July 2009

In August 2009, the LSE introduced a second dynamic tick size regime (cf. Table B.8) with effect from 21 September 2009 to enhance price granularity for most liquid FTSE 100 stocks. The stocks submitted to this regime moved to a new market segment, “SET0”, created for the purpose. At the time of its introduction, the SET0 tick size regime was applied to the 12 FTSE 100 securities listed in Table B.9.

Table B.8

Price (in GBX/GBP/USD/EUR) from	To	Tick size
0	0.9999	0.0001
1	4.9995	0.0005
5	9.999	0.001
10	49.995	0.005
50	99.99	0.01
100	499.95	0.05
500	999.9	0.1
1,000	4,999.5	0.5
5,000	9,999	1
10,000		5

Source: LSE Service Announcement 001/180809 - 18 August 2009.

Table B.9

Stock name	ISIN code	Implementation date
ANGLO AMERICAN	GB00B1XZS820	21 September 2009
ASTRAZENECA	GB0009895292	
BARCLAYS	GB0031348658	
BHP BILLITON	GB0000566504	
BP	GB0007980591	
BT GROUP	GB0030913577	
GLAXOSMITHKLINE	GB0009252882	
HSBC HLDGS	GB0005405286	
LLOYDS GRP.	GB0008706128	
ROYAL DUTCH SHELL ‘A’	GB00B03MLX29	
RIO TINTO	GB0007188757	
VODAFONE GROUP	GB00B16GWD56	

Source: LSE Service Announcement 001/180809 - 18 August 2009.